26-S98-3

IE 4598 Final Report

Project #3: Greater Baton Rouge Food Bank Routing Optimization For Deliveries

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5/6/2026

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Executive Summary

This project aimed to optimize the delivery scheduling and routing operations of the Greater Baton Rouge Food Bank to improve efficiency, reduce fuel consumption, and create a sustainable scheduling framework. The food bank currently serves over 116 agencies with varying delivery frequencies and strict volunteer time constraints, leading to inefficient routes caused by agency preferences rather than operation optimization.

Key findings indicate that inefficiencies are caused by unstandardized processes, reliance on manual data handling, and underutilization of Samsara, the food bank's current fleet management software. Analysis confirmed that Samsara's routing tool produces near-optimal routes, averaging only 0.35 miles above the optimal route, validating its reliability for continued use.

The team has developed several recommendations to improve both the routing and data processes. This includes visual clustering of agencies using Google MyMaps, digitizing the current paper-based systems with structured Excel tools, pivot tables, an Electronic Document Library, and a Google Form to streamline scheduling and improve data accuracy.

Implementation efforts revealed significant improvements for the food bank. The Pivot Tool was able to reduce schedule creation time of each month from approximately 67 minutes to 45 minutes. Projected estimation of results after clustering agencies reduced the total mileage by 18-22% per month when compared to the original requests. The clustering approach translated into substantial cost reductions and environmental benefits, including a potential gas savings of ~$4200 and an approximate reduction of 2 metric tons of carbon dioxide over a two-month period.

Overall, this project delivers a scalable and repeatable optimization framework that improves operational efficiency and data reliability, reduces fuel costs, and supports long-term sustainability goals. By combining process standardization, digital tools, and route optimization, the Greater Baton Rouge Food Bank is able to significantly enhance its delivery operations in the coming years.

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Background and Motivation

The Greater Baton Rouge Food Bank is a 501(c)(3) nonprofit organization that procures, inventories, warehouses, and delivers donated food. In 2024, there were 12.8 million meals (16.6 million pounds) distributed across the 11-parish area to various pantries, meal sites, homes, shelters, and soup kitchens.

Up to 1.3 million pounds of food are sent to the Greater Baton Rouge facility each month, with each client receiving 15 or more pounds of fresh produce, canned goods, frozen meats, and dry products. The agencies set distribution times, the number of visits each month, and drop off locations. The food bank is expected to deliver to all 116 agencies on a weekly, biweekly, or monthly basis, illustrating the need for an effective routing system.

Problem Description

Since its founding, the Greater Baton Rouge Food Bank has catered to the agencies' requests for specific days and times to receive deliveries, due to the schedules of the volunteers. The routing has been crafted to accommodate these agencies and their volunteers, leading to suboptimal routes.

Rising fuel costs and federal budget cuts have increased the need to optimize routing and vehicle use. Idle time has also increased due to increased traffic and inconsistent driving patterns, frequently occurring when drivers deviate from the scheduled routes. The drivers and management are also not fully relying on Samsara (the current fleet management software), resulting in inaccurate data reports and inefficient use. When the employees are not following the schedule and not updating the routes to reflect their actual paths, the data stored in the software becomes unreliable.

Management Objectives

The original goal of the study was to analyze current transportation operations (routing, scheduling, and fuel) and implement fuel-efficiency improvements to support ongoing transportation optimization. The team aimed to maximize cost savings by reducing fuel consumption by 5-10%, improving delivery efficiency, and ensuring all delivery needs are met within the Baton Rouge community.

However, due to the nature of the food bank's operations, the objectives shifted in the second half of the project. With many agencies being unable to change their delivery days without a year's notice, the team shifted focus towards creating an optimization strategy for the food bank to implement in the upcoming 2027 operational year. Management was also interested in digitizing the current manual process of creating the operational schedule and storing data.

Management had requested that the team create an optimization strategy that can be continued year after year. The team first built a heuristic tool for schedule building, verified Samsara's route optimization capabilities, and built digital monitoring tools through Power BI. To further optimize the delivery scheduling system, the team organized agency move requests

and location data to create a universal tool for future location-based delivery planning. Overall, management requested that the team create a framework for the schedule-building procedure to implement in the future.

Data Collection

Samsara

[AS]The core focus of this study is to identify current issues within the food bank transportation system, including driver idling, unnecessary distance in routing, and adherence to client schedules. The team has collected metrics based on driver traveling trends to serve as a baseline for future improvement measurement.

The data utilized for baseline metrics of this project was collected from the period of January 1st 2025 to May 31st 2025. The specific data for this period is currently stored within the Samsara platform, and it was exported for use. Vehicle metrics were summarized in Table 1. The improved goal metrics were calculated by taking 95% of the current metrics, resulting in a 5% reduction of each measurement except for fuel efficiency, which should increase by 5%. For future comparisons, the food bank will use exports from Samsara to compare metrics over time. The indicators being considered include efficiency, fuel used, distance, and idle time as highlighted in the table below.

Table 1: Baseline performance indicators, compiled from Samsara.

Vehicle	Efficiency (MPG)	Fuel Used (gal)	Distance (mi)	Est. Carbon Emissions (kg)	Est. Cost	Total Engine Run Time (min)	Idle Time (min)	Idle Time (%)
Tractor 2/P	6.38	741.71	4730.42	7572.29	2711.06	13320.01	5999.01	45.04
Truck 1	9.37	797.09	7469.93	8137.71	2862	14713	1219.35	8.29
Truck 7	8.64	437.17	3775.15	4463.21	1590.58	7315.87	449.09	6.14
Truck 5	8.52	682.97	5818.87	6972.59	2463.76	13379.08	3407.06	25.47
Tractor 1/R	7.05	689.83	4861.45	7042.64	2455.42	14869.4	5509.48	37.05
Truck 3	8.39	1036.68	8694.57	10583.74	3722.68	18510.92	3868.44	20.9
Truck 2	9.53	458.83	4373.08	4684.34	1626.79	10786.86	2558.13	23.72
Truck 4	5.04	5.94	29.97	60.67	20.81	248.84	170.21	68.4
Average	7.865	606.2775	4969.18	6189.64875	2181.6375	11642.9975	2897.59625	29.37625
Goal Average (based on improvement	**8.26**	**575.96**	**4720.72**				**2752.72**	**27.91**

Originally, the selected performance metrics of efficiency, fuel usage, distance, and idle time were set to help the team identify if optimized routes would have a positive impact on the goal. However, the inability to shift routes in 2026 restricted data comparison. An overall understanding of how to collect and analyze data from Samsara will benefit the food bank in future years as they work towards key metric reductions, especially fuel consumption.

The data can also provide clarity into the specific decisions made by drivers. The GBRFB wants to ensure that deliveries are on time and meet the internal standard of 30-minute unloading. Drivers often deviate from routes based on convenience, such as dropping off a shipment before its scheduled time, knowing an individual would be there to receive it. This can cause issues with not only the data, but also agency relations, since the agencies are heavily reliant on volunteers and therefore their availability. It can be difficult for an elderly person to bring in a shipment by themself if it was left for them to deal with alone. There are also long wait times for drivers when they deliver to distribution centers because they must wait for all palettes to be unloaded from the truck individually. By evaluating where there are inconsistencies in the Samsara data, the food bank can work towards setting specific goals for observed metrics, leading to a more efficient delivery system using visualized data.

Agency Calendars

[RS]One struggle of the Greater Baton Rouge Food Bank is data organization, especially when it comes to calendar creation for deliveries and pickups. There is currently a master Excel sheet that holds all agency data, pickup and delivery statuses, points of contact, and addresses. The team's new data sheet was created through manual entry of agency information, sourced from printed documents. The food bank relies on a physical data storage system, which is a legal requirement due to its nonprofit status.

While having physical records is a requirement, sorting through the sheer number of documents leads to mistakes and excess processing time. To increase accuracy and minimize manual searching time when scheduling deliveries, a Client Records Excel sheet was created. The GBRFB forwarded all PDF documents of agency requests, and all requests were recorded per agency based on a weekly, biweekly, or monthly delivery frequency.

The team sliced mass PDFs into individual agency calendars and then took detailed notes on delivery requests for every month of 2026 for each agency (Table 2). Agencies were categorized by their delivery frequency and grouped by the main delivery day requested. Some agencies had one-off requests, where one delivery of the month differed from the normal delivery trend, mostly due to GBRFB closures. These alternate days were recorded in a separate column, for organizational purposes. Using lookup functions in Excel, agency request days, frequencies, and notes were pulled into the main raw data file.

Table 2: Excel data collection sheet of all agency information.

Delivery Point	Freq.	Notes on Days requested	Main days	Alt Days
Alsen St. Irma Lee Community Village	Monthly	Third Tuesday of the month (Feburary 2nd Tuesday)	Tues	
Deaf Resource Center	Biweekly	Tuesdays (only 2 deliveries per month, start with first tuesday) [F	Tues	
East Feliciana Council on Aging	Monthly	Second Tuesday of the Month	Tues	
Frank Sotile Pavillion	Monthly	Second Tuesday of every month	Tues	

By associating agencies with their requested delivery frequencies and days, the user can quickly filter by day requested to create the initial layout of the delivery calendar. The food bank staff have received a Standard Operating Procedure (SOP) for continuing online data collection to create this sheet (Appendix: SOP-002: Google Form). The SOP-002 details how to use a Google Form to record information, and then how to apply Excel functions to organize the data in an easy-to-read sheet. The SOP-001: Client Record Pivot Tool details how to copy the information over into Excel for further analysis (refer to Client Record Pivot Tool under Data Analysis and Findings for further analysis).

Truck Audits and Driver Surveys

[MP]Driver input was gathered using opinion-based surveys completed by six drivers in the Greater Baton Rouge Food Bank fleet (Figure 1). The truck numbers shown in the table below simply represent the vehicles assigned to each driver and were used for identification. Drivers rated their overall route efficiency on a scale from 1 to 10, reported how often they experienced delays, and provided suggestions for improvements based on their day-to-day experience (Table 3).

Name:

Truck #:

Section 1: Background Information

1. How long have you been driving for the Food Bank?
   - ☐ Less than 6 months
   - ☐ 6–12 months
   - ☐ 1–3 years
   - ☐ More than 3 years
2. On average, how many hours per week do you spend on Food Bank deliveries?
   - ☐ Less than 10 hours
   - ☐ 10–20 hours
   - ☐ 21–30 hours
   - ☐ More than 3 hours

Section 2: Delivery Challenges

3. How often do you encounter delays when delivering to partner agencies?
   - ☐ Rarely
   - ☐ Sometimes
   - ☐ Frequently
   - ☐ Almost always
4. What are the main causes of delays in Food Bank deliveries? *(Select all that apply)*
   - ☐ Traffic congestion
   - ☐ Poor or unclear routing/directions
   - ☐ Vehicle issues
   - ☐ Loading/unloading times at Food Bank warehouse
   - ☐ Loading/unloading times at agency sites

Figure 1: Truck Driver Survey.

Table 3: Driver Survey Results.

Truck #	Efficiency Rating (1-10)	Delay Frequency	Improvement Suggestion
6	10	Rarely	Automatic Rerouting
916	7	Sometimes	Better Agency Scheduling
3	9	Rarely	None
5	8	Rarely	Better Agency Scheduling
2	2	Frequently	Better Agency Scheduling
439260	7	Frequently	Better Agency Scheduling
Average	**7.17**		

Truck 6 gave the highest efficiency score of 10 and reported that delays rarely occur, with automatic rerouting listed as the only suggested improvement. Truck 3 also performed well with a rating of 9, rare delays, and no recommended changes. Truck 5 earned a rating of 8 with rare delays, while Trucks 916 and 439260 both received ratings of 7. These drivers reported delays occurring *sometimes* to *frequently* and consistently suggested better scheduling with agencies as the main improvement. Truck 2 gave the lowest efficiency rating of 2 and reported

frequent delays, showing significant challenges with current scheduling. Better coordination with agencies was again identified as the needed solution. The overall average efficiency rating across all surveyed drivers was 7.17, which indicates that most drivers felt their routes were generally efficient, but considering the one outlier, improvements can still be made. It was determined that scheduling with agencies remains the main source of issues, which validated the goal of routing agencies based on location.

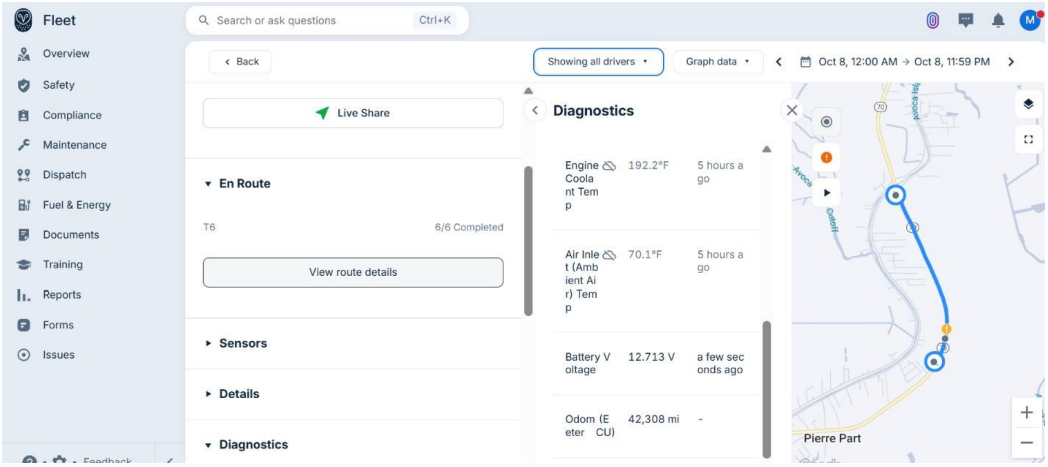


Figure 2: Samsara Dashboard.

The team audited three different routes, with one of the routes having two team members due to the lack of deliveries that day. During these sessions, the following metrics were manually recorded: start and end time of the route, arrival and departure times of each stop, mileage and gas usage from the odometer, and idle times. The team also noted any deviations from the planned routes, instances of distracted driving, and the software usage on the mobile interface. After completing the field observations, the team went back into Samsara to compare the manually recorded values to the software-tracked values. Figure 2 visualizes how route details were collected from the system. Analyzing these numbers allowed the team to conclude that Samsara's data tracking system is quite accurate (Table 4). The slight deviations in numbers can be attributed to drivers stopping to refuel and certain road closures or restrictions during the route. While the tracking technology functions effectively, improvements in driver communication, SOPs, and data maintenance would further improve efficiency.

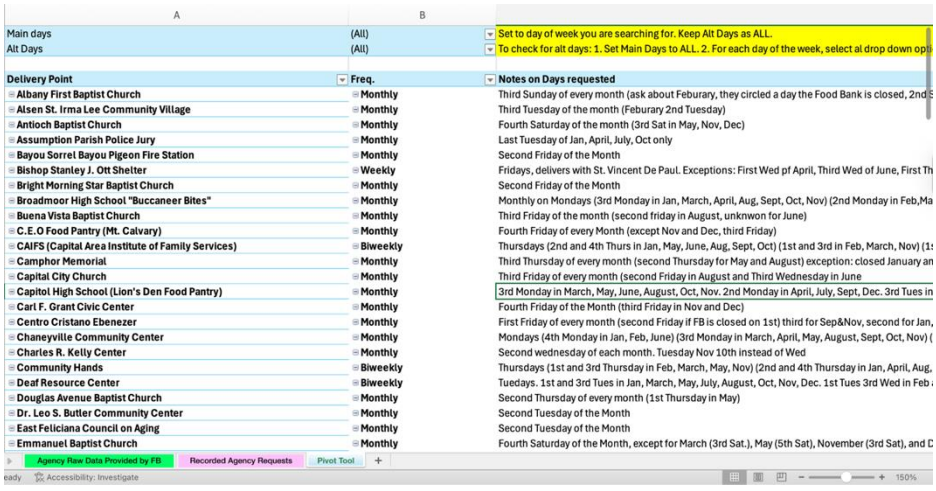
Table 4: Truck audit and Samsara results.

Route	***Date***	***Departure Time (Manual)***	***Departure Time (Samsara)***	***Return Time (Manual)***	***Return Time (Samsara)***	***Mileage (Manual)***	***Mileage (Samsara)***
1	10/08/2025	6:49 AM	6:49 AM	1:16 PM	1:16 PM	165	164
2	10/10/2025	6:36 AM	6:36 AM	8:24 AM	8:23 AM	43	43.09
3	10/10/2025	7:11 AM	7:11 AM	9:56 AM	9:56 AM	137.6	132

SOP-005, within the Standard Operating Procedures section of the report, outlines what daily tasks at the Greater Baton Rouge Food Bank should look like for drivers and management. This procedure was created to ensure that Samsara is working effectively and has as few human errors as possible. It also outlines what drivers should do in case of setbacks during deliveries so that these shortcomings are accurately tracked for future avoidance.

Data Analysis & Findings

[RS] Once all agency requests were recorded, sorted, and pulled into the main data sheet, the Client Records Pivot Tool was created to include all essential data for the process of calendar creation (Figure 3). The filters at the top of the pivot allow the user to sort by day of the week. By selecting one day in particular, all weekly, biweekly, and monthly deliveries that occur on that day within the month are visible. The agency's physical address and contact information associated with the delivery are also accessible by pulling alternative fields into the Pivot Tool, which is essential for later organization of deliveries by location. SOP-001: Client Record Pivot Tool explicitly details all steps for using the Pivot Tool for efficient delivery scheduling. To locate any off-trend deliveries, the Main Days filter can be set to ALL, while the alternative delivery day is set to the day in question. The notes attached to each agency request are detailed, and the user can quickly see if the specific month being sorted is mentioned in the notes as having any off-trend deliveries.



	A	B	
Main days		(All)	Set to day of week you are searching for. Keep Alt Days as ALL.
Alt Days		(All)	To check for alt days: 1. Set Main Days to ALL. 2. For each day of the week, select al drop down op
Delivery Point		Freq.	Notes on Days requested
Albany First Baptist Church		Monthly	Third Sunday of every month (ask about Feburary, they circled a day the Food Bank is closed, 2nd S
Alsen St. Irma Lee Community Village		Monthly	Third Tuesday of the month (Feburary 2nd Tuesday)
Antioch Baptist Church		Monthly	Fourth Saturday of the month (3rd Sat in May, Nov, Dec)
Assumption Parish Police Jury		Monthly	Last Tuesday of Jan, April, July, Oct only
Bayou Sorrel Bayou Pigeon Fire Station		Monthly	Second Friday of the Month
Bishop Stanley J. Ott Shelter		Weekly	Fridays, delivers with St. Vincent De Paul. Exceptions: First Wed pf April, Third Wed of June, First Th
Bright Morning Star Baptist Church		Monthly	Second Friday of the Month
Broadmoor High School "Buccaneer Bites"		Monthly	Monthly on Mondays (3rd Monday in Jan, March, April, Aug, Sept, Oct, Nov) (2nd Monday in Feb,Ma
Buena Vista Baptist Church		Monthly	Third Friday of the month (second friday in August, unknwon for June)
C.E.O Food Pantry (Mt. Calvary)		Monthly	Fourth Friday of every Month (except Nov and Dec, third Friday)
CAIFS (Capital Area Institute of Family Services)		Biweekly	Thursdays (2nd and 4th Thurs in Jan, May, June, Aug, Sept, Oct) (1st and 3rd in Feb, March, Nov) (1
Camphor Memorial		Monthly	Third Thursday of every month (second Thursday for May and August) exception: closed January an
Capital City Church		Monthly	Third Friday of every month (second Friday in August and Third Wednesday in June
Capitol High School (Lion's Den Food Pantry)		Monthly	3rd Monday in March, May, June, August, Oct, Nov. 2nd Monday in April, July, Sept, Dec. 3rd Tues in
Carl F. Grant Civic Center		Monthly	Fourth Friday of the Month (third Friday in Nov and Dec)
Centro Cristano Ebenezer		Monthly	First Friday of every month (second Friday if FB is closed on 1st) third for Sep&Nov, second for Jan,
Chaneyville Community Center		Monthly	Mondays (4th Monday in Jan, Feb, June) (3rd Monday in March, April, May, August, Sept, Oct, Nov) (
Charles R. Kelly Center		Monthly	Second wednesday of each month. Tuesday Nov 10th instead of Wed
Community Hands		Biweekly	Thursdays (1st and 3rd Thursday in Feb, March, May, Nov) (2nd and 4th Thursday in Jan, April, Aug,
Deaf Resource Center		Biweekly	Tuedays. 1st and 3rd Tues in Jan, March, May, July, August, Oct, Nov, Dec. 1st Tues 3rd Wed in Feb
Douglas Avenue Baptist Church		Monthly	Second Thursday of every month (1st Thursday in May)
Dr. Leo S. Butler Community Center		Monthly	Second Tuesday of the Month
East Feliciana Council on Aging		Monthly	Second Tuesday of the Month
Emmanuel Baptist Church		Monthly	Fourth Saturday of the Month, except for March (3rd Sat.), May (5th Sat), November (3rd Sat), and D

Agency Raw Data Provided by FB | Recorded Agency Requests | Pivot Tool | +

eady | Accessibility: Investigate | 150%

Figure 3: Final Pivot Tool for calendar creation.

By consolidating delivery frequencies and days for requests within the master datasheet, visuals were created in Excel to show the distribution of delivery frequencies and specific days requested over the 2026 year. By visualizing delivery request trends, the team had clarity as to which specific days and types of deliveries were most frequent. Using this information, the team suggested move requests that were applicable and most likely doable by partner agencies, to work towards future mileage reduction in travel.

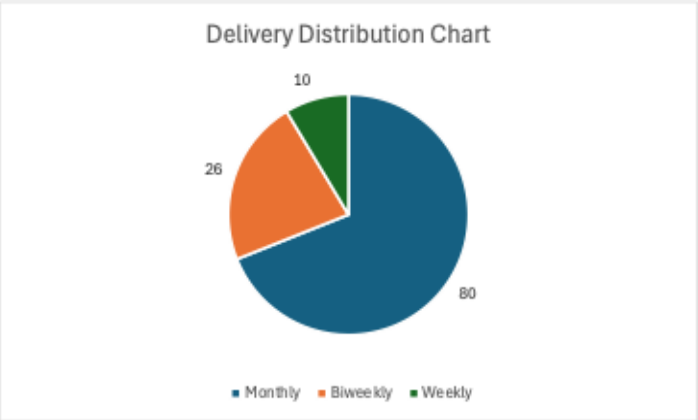


Figure 4: Pie chart of delivery frequencies

Most deliveries occur on a monthly basis, with the least number of deliveries being weekly (Figure 4). Through speaking with upper management on agency trends, it was noted that agencies request specific delivery days because they can ensure volunteer assistance on those days. By understanding the nature of agency behaviors and the spread of the data, the team could assume that agencies with monthly deliveries would be more receptive to accepting different delivery dates if location clustering led to a change of delivery day. These agencies only need volunteer assistance one day of the month, which would be easier to re-coordinate than changing the schedule of volunteers who come weekly. By knowing which agencies may have the capacity to adapt to adjustments, the team had more freedom in recommending schedule patterns that meet the requirements of the GBRFB.

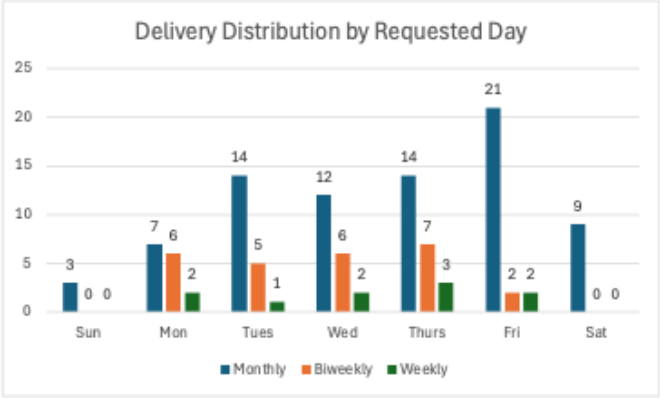


Figure 5: Bar chart of agency requests, by day

The distribution of main delivery days requested was also essential to visualize, so that the data can be fully understood (Figure 5). Currently, most deliveries occur on Fridays, with Thursdays and Tuesdays also having a high number of requests. By visualizing which days had the most deliveries, the team had a greater opportunity for delivery grouping on these days, especially if delivery points were within a close distance of each other.

Recommendations

[SD]Samsara has an "optimize route" feature (Figure 6), but there is a maximum of 20 stops. While the team has suggested an optimization strategy that first groups agencies by location zones before Samsara implementation, this feature can aid in further mileage reductions. Samsara analyzes which sequence of stops has the shortest distance and factors in expected traffic conditions for the predicted time of day to determine travel time. Since the food bank will also be using Samsara moving forward, the team incorporated full reliance on Samsara's AI route optimization into the final solution (Reference Statistical Analysis: Samsara Optimal Route Creation to understand the optimality of Samsara route creation).

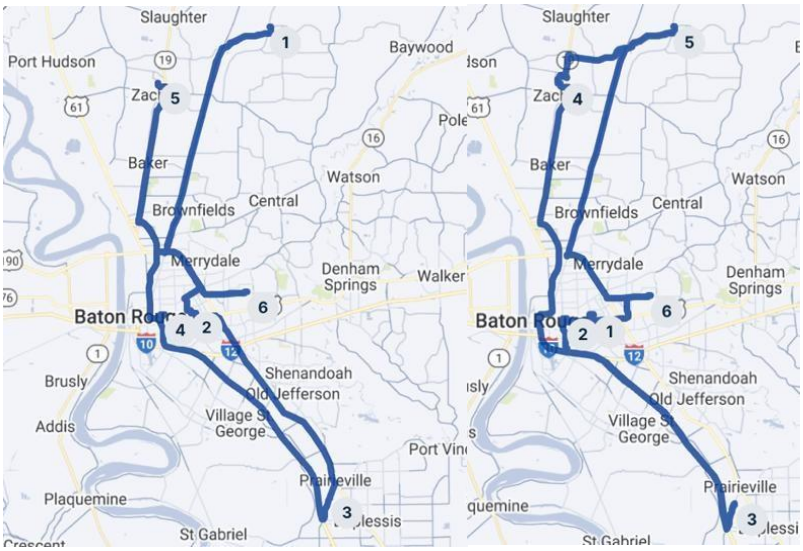


Figure 6: Samsara Optimize Route feature (Before and After).

Research has been collected on a Google Maps feature called MyMaps. This software allows the user to tag locations to visually see their placement. Upon further investigation, it was found that a CSV file can be imported into this software; the column titles include the name, address, latitude, longitude, and identifier. Using the address book CSV file extracted from Samsara, containing all of the agencies, their addresses, latitude, longitude, and other values, all of the agencies that requested deliveries in the first week were compiled into a CSV file (Table 5).

Table 5: Excel visual of the sample CSV file of agency inputs.

Name	Address	Latitude	Longitude	Day of Week
O'Dell Tickles Food Pantry	24497 Cook Rd, Zachary, LA, 70777	30.7086142	-91.041613	Monday
Resurrection Life Ministries	722 N Carrollton Ave, Baton Rouge, LA 70806, USA	30.455389	-91.123026	Monday
St. John Outreach Pantry	15208 Hwy 73, Prairieville, LA 70769	30.2908193	-90.977077	Monday
True Light Multi Center	3836 North Street, Baton Rouge, LA, 70806	30.4532584	-91.151178	Monday
Zachary Food Pantry	5640-A Rollins Rd, Zachary, LA, 70791	30.6587059	-91.156996	Monday
Deaf Resource Center	16288 Tiger bend RD	30.3884586	-91.004074	Tuesday
Gumbo	23440 Railroad Ave, Plaquemine, LA 70764, USA	30.2901125	-91.236775	Tuesday
Holy Grill	3112 Convention St/6117 Cadillac St	30.4489357	-91.156543	Tuesday
Holy Rosary	44450 LA-429, St Amant, LA 70774, USA	30.2257599	-90.856122	Tuesday

This information was then uploaded into MyMaps, producing the results below (Figure 7). This creates a visual cluster by location of each agency, and the colors indicate the day of the week the agency requested. To have the most optimal routes, some agencies will have to have their deliveries shifted to a different day of the week, based on what the majority of agencies in the vicinity requested. This alternative is feasible for the food bank, as the software can handle up to 2000 rows and is completely free to use.

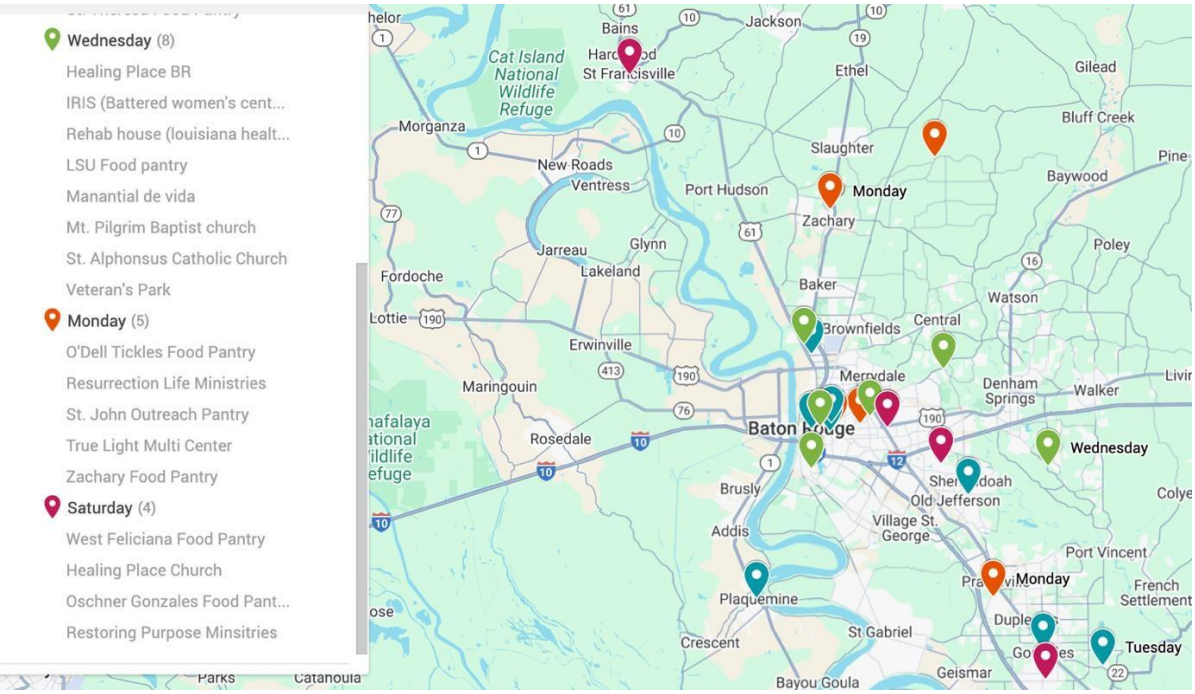


Figure 7: MyMaps CSV visual mapping output.

Original MyMaps Routing Strategy

[SD]MyMaps is a Google Maps program that allows users to create and share custom maps. These maps do not give routing suggestions but rather lay out all addresses spatially. MyMaps can filter datapoints by assigning colors and labels to specific locations. This tool was heavily utilized by the team to divide route assignments and determine optimal adjustments to current routes.

The first version of the operational schedule consisted of the team using the Client Records Pivot Tool to assign deliveries to the requested day of the month. The deliveries were not assigned to trucks or to timeframes initially. After collecting one month of delivery addresses and requested days, each agency name was recorded in an Excel sheet to create the file for MyMaps implementation. MyMaps allows for mass uploads of addresses, but they must be in CSV format and include the agency address, latitude, and longitude, as well as the identifying data points of agency name and day of the week requested.

After pulling in Agency names, addresses, coordinates, and requested days, the month was downloaded in CSV format and uploaded to MyMaps. Once within MyMaps, the locations were color-coded by day of the week.

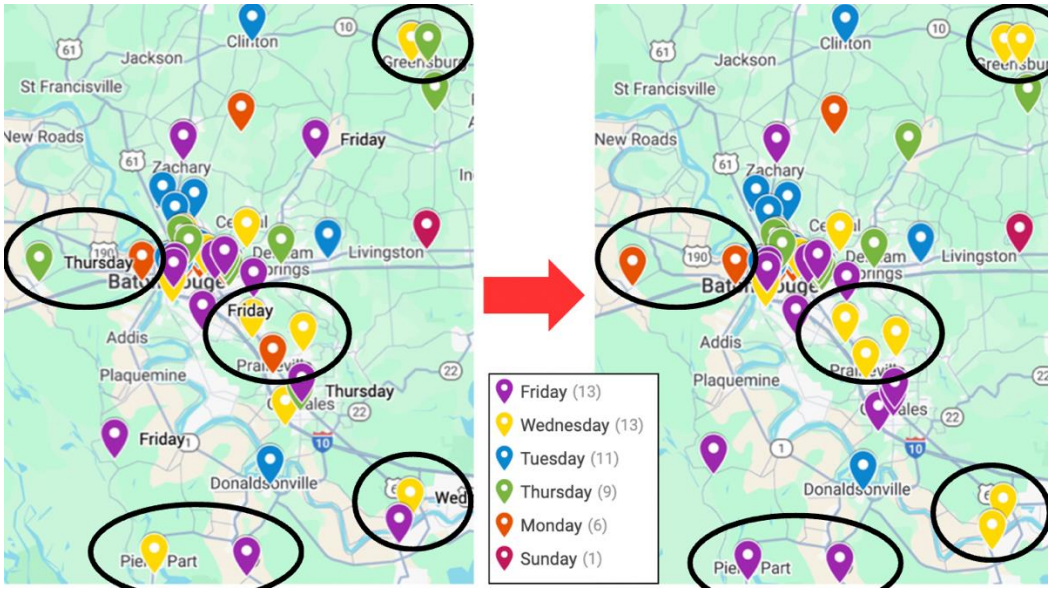


Figure 8: MyMaps visual clustering (before and after).

Figure 8 above is a sample of a week in February. On the left, the agencies were input as requested, with each day of the week having a specific color. Orange for Monday, blue for Tuesday, yellow for Wednesday, green for Thursday, purple for Friday, and magenta for Saturday. The team first created an operational schedule based solely on the agency requests. This schedule served as a baseline for economic comparison in terms of original mileage and fuel spend, but due to the inability of agencies to adjust their deliveries, the food bank used this schedule as its main operational schedule.

The rearranged operational calendar was then created using visual clustering and the baseline requests on MyMaps. The team analyzed the radius from one agency location to another, and if there were similarities with the path (connected with the same interstate), that would make for a more convenient delivery route. Based on the original map, it was clear that some agencies should be delivered on different days, due to the presence of other deliveries in the surrounding area. However, mainly agencies with monthly requests were moved because an agency with a weekly delivery request is unlikely to accept a different delivery day for every week. With monthly requests, asking that the agency move to a different day of the week would be more feasible. Since the team wants this rearrangement to be as efficient as possible, the focus remained on the monthly deliveries for a higher approval success rate. After noting the move requests, the CSV file was edited and reinserted into MyMaps, creating the image on the right of Figure 8. Visually, this output reduced the amount of backtracking amongst the days of the week. Week by week, the team made move suggestions for the months of January, February, and March. These moves served as the basis for the strategic zoning plan under Implementation. The original move requests also served as the comparison factors for economic analysis in the Economic and Environmental Justification section.

Implementation

Updated MyMaps Strategy: Strategic Zoning

[SD]While creating maps for each week of each month was beneficial in recognizing patterns and visually clustering, this process was too time-consuming to expect the food bank to replicate in the future. As a more efficient approach, a MyMaps visual was created to show 7 zones, based on k-means clustering (Figure 9). A k-value of 7 was used to represent the constraint of 7 trucks. This map was intended to provide the food bank with a visual of the zones that have a high impact on mileage, which will aid them in creating delivery routes. Within this map, there were sheets that showed the original requested days for each zone, so that the food bank could see the current layout of requests (Figure 10). The clustering was not capacity-constrained due to varying pallet weights and sizes each week, as donations to the food bank change based on donors' available surplus.

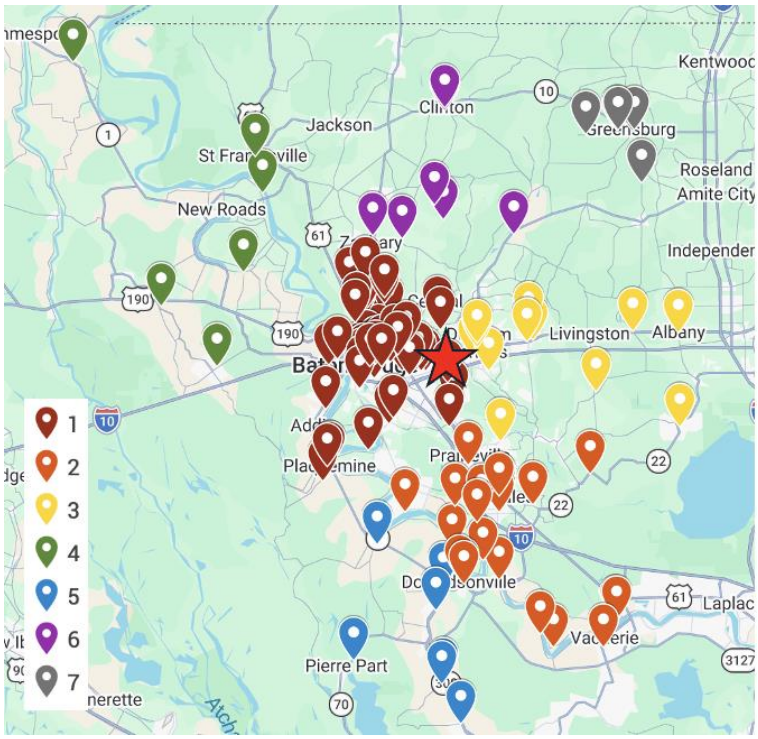


Figure 9: Agency zones; red star Marks GBRFB location.

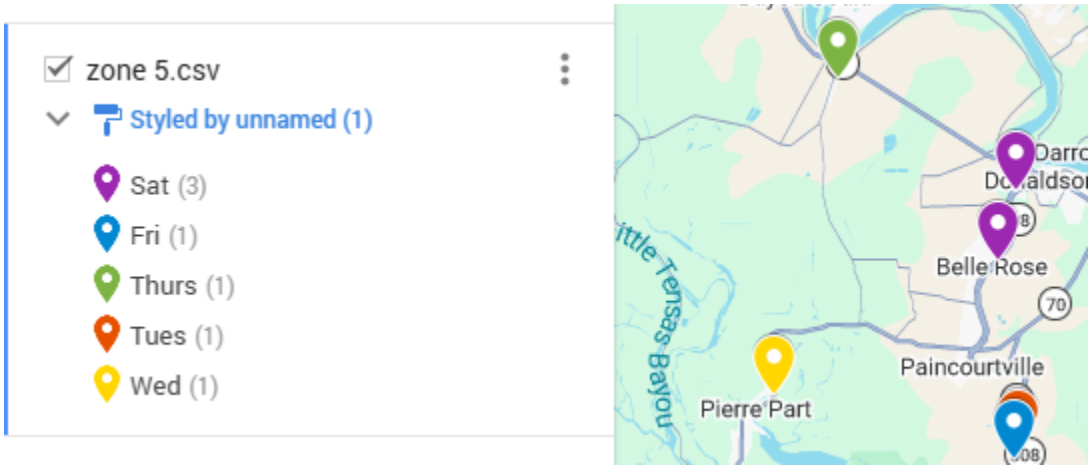


Figure 10: MyMaps of Zone 5.

[RS]The team compiled a list of all agencies that were recommended to be delivered on the same day via the same truck, based on visual clustering of requests in January, February, and March. 75 agencies were grouped after performing this original location clustering. The team found that 25% (19) of agencies moved were due to strategic grouping, meaning that the locations were located on the same major roadways, but not necessarily in the same zone. 75% (56) of the agencies that were clustered were in the same zone, with zone 4 not having any clusters. Figure 11 depicts which zones were most impacted by visual clustering. Zones 1 and 2 are East Baton Rouge and Ascension, respectively.

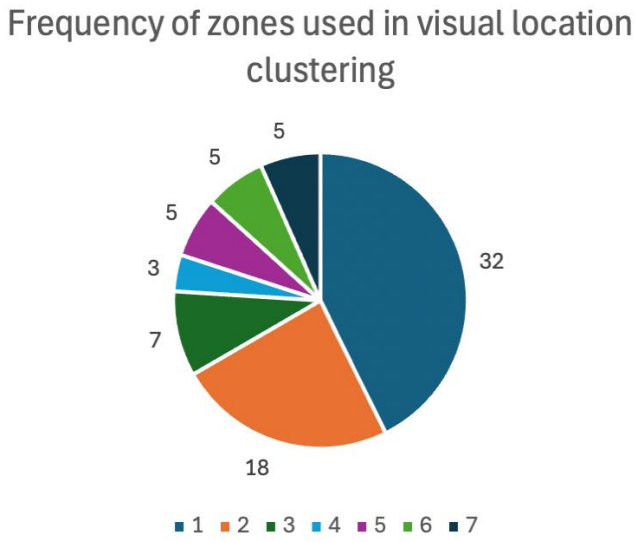


Figure 11: Pie chart of zone frequency noted in visual location clustering.

Visual clustering and clustering by zoning have proved to be similar in their result, as a majority of move requests fell within the same zones. By implementing the zoning strategy, the food bank can work towards reducing its mileage and creating the most optimal routes.

[SD]The team implemented the zoning strategy by producing all necessary MyMaps visuals and finalizing a list of suggested delivery changes. This information was given to the

GBRFB employees for their future use. This new strategy of grouping agencies by zone will be less intricate and easier for the food bank to apply across all agencies, compared to the original strategy of analyzing every week.

Digitalizing Process & Data Standardization

[AS]An additional recommendation for this improvement project included an electronic document library, allowing for increased organization and efficiency. Currently, the food bank sends out paper calendars to agencies for them to mark their preferred days, scanning the results back into one giant PDF, and keeping the paper copies in a folder. The current employee who is in charge of creating the schedule expressed her frustration with flipping through the massive amount of agency calendars, since the binder contains 2 years' worth of information. The large PDF file has been separated for each agency, allowing the users to find the desired agency quickly. After naming each file to correspond with its agency and compiling them into one folder, employees can easily use the "ctrl+f" keyboard shortcut to search through the folder. Now, all files are stored in one place where they can be accessed easily (Figure 12).

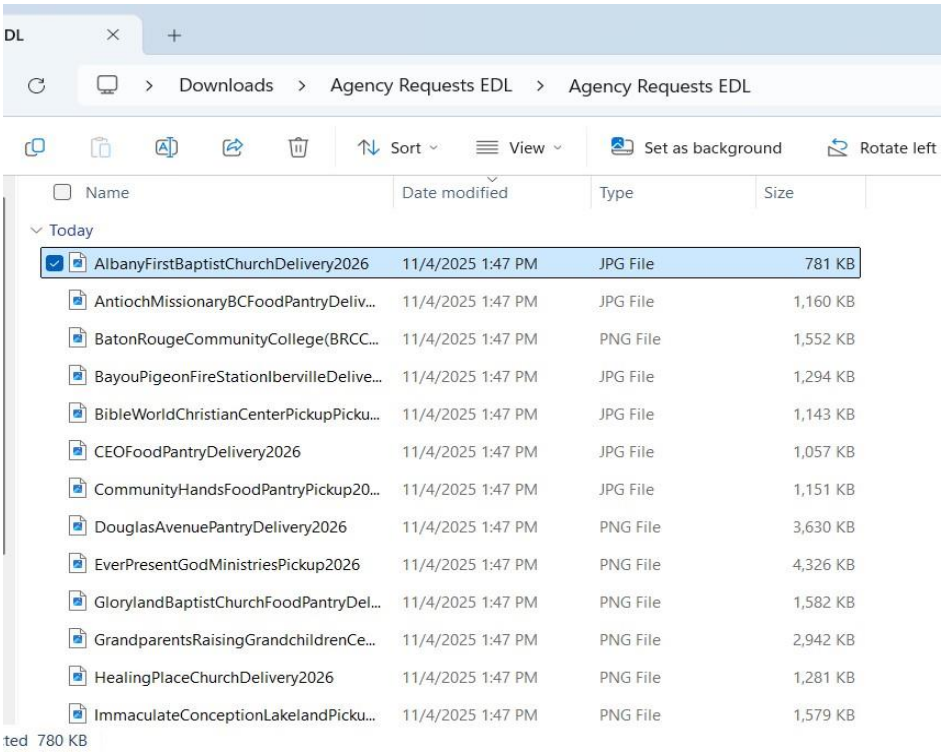


Figure 12: Alphabetized Electronic Document Library for agency calendars.

[SD]The integration of Google Forms for data collection is another step in digitizing agency information (Figure 13). The curated form collects the name of the agency, and using IFERROR and VLOOKUP functions, the response sheet will automatically record the corresponding delivery address. The food bank currently asks agencies to fill out a paper request calendar, which serves as the basis of the team's EDL. To ensure legal requirements are met, the food bank will still ask agencies to continue filling out the paper request, but then scan and upload it to the Google Form. Instructions have been created for this process, detailing the steps

to edit the form and extract the responses (SOP-002: Google Form). While the Google Form will be the creation point for making future EDLs, the Google Form also serves the purpose of collecting delivery details, so the food bank can easily recreate the Pivot Tool year after year (Part 2 of SOP-001: Client Record Pivot Tool). The form prompts the user to fill out the requested frequency of deliveries (weekly, biweekly, or monthly), notes corresponding to each delivery, and the main and alternative days of delivery being requested. Once submitted, food bank staff can add additional notes and customize them to be as detailed as necessary. The Google Form can be implemented instantly, as the food bank can begin utilizing the tool internally to easily digitize current physical data.

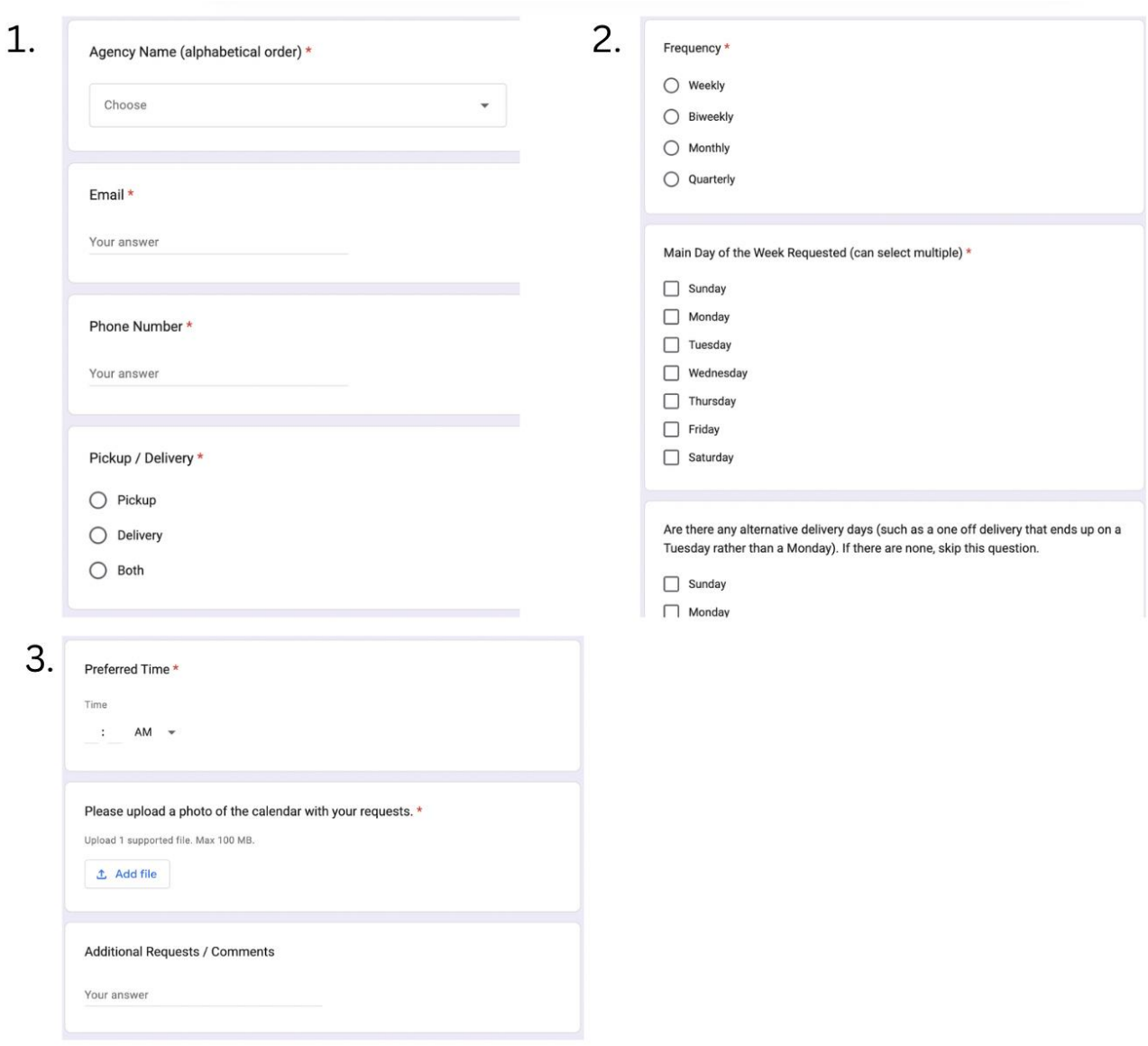


Figure 13: Sample Google Form for agency requests.

[RS, SD]After all the agencies submit the form, the food bank will be able to extract the responses into a spreadsheet, shown below in Table 6. Google Sheets' filters allow the food bank

to sort the responses. The dropdown option for the Agency Name allows the food bank to connect each agency to its respective address by using VLOOKUP functions, as shown in Table 7. The IFERROR functions are used to pull response data into the new sheet, which visually cleans the data and makes it easier to copy and paste into Excel for analysis. The functions pull data from the Address Book and Zones sheet, shown in Table 8. With this, the food bank can recreate the Pivot Tool.

Table 6: Sample response spreadsheet of agency requests.

Agency Name (alph	Email	Phone Number	Pickup / Delivery	Frequency	Main Day of the We	Preferred Time	Please upload a photo of the calendar with yc
Albany First Baptist Chu	email@gmail.com	1234567890	Delivery	Weekly	Sunday	8:00:00 AM	https://drive.google.com/open?id=1Tj97nh6tC3fk
Alsen St. Irma Lee Comr	email@gmail.com	1234567890	Delivery	Biweekly	Tuesday, Thursday	9:00:00 AM	https://drive.google.com/open?id=1xypdvou2-o_a
Bishop Stanley J. Ott Sh	email@email.com	1234567890	Delivery	Monthly	Sunday, Thursday	10:00:00 AM	https://drive.google.com/open?id=1vKyFxjgHXrw
C.E.O Food Pantry (Mt. (	email@email.com	1234567890	Delivery	Biweekly	Wednesday	7:00:00 AM	https://drive.google.com/open?id=1gX24gHMLHS

Table 7: Responses with addresses.

=IFERROR(VLOOKUP(A2,'Address Book and Zones'!A:C,3,FALSE), "")

Agency Name	Address	Email	Phone Number	Pickup/Delivery	Frequency	Main Day	Alt Day	Comments	Preferred Time
Albany First Baptist Church	18950 Florida Blvd.	email@gmail.com	1234567890	Delivery	Weekly	Sunday	Monday		8:00:00 AM
Alsen St. Irma Lee Community Village	601 Old Rafe Mayer Rd.	email@gmail.com	1234567890	Delivery	Biweekly	Tuesday, Thursday		I have requests t	9:00:00 AM
Bishop Stanley J. Ott Shelter	1623 Convention St.	email@email.com	1234567890	Delivery	Monthly	Sunday, Thursday			10:00:00 AM
C.E.O Food Pantry (Mt. Calvary)	6433 Highway 405	email@email.com	1234567890	Delivery	Biweekly	Wednesday	Thursday		7:00:00 AM

Table 8: Address Book and Zones.

	Agency Name	Parish	Address 1	City	Zip	Latitude	Longitude	Zone
2	Albany First Baptist Church	Livingston	18950 Florida Blvd.	Albany	70711	30.5035921	-90.586215	3
3	Alsen St. Irma Lee Community Village	East Baton Rouge	601 Old Rafe Mayer Rd.	Baton Rouge	70807	30.572207	-91.1993785	1
4	Antioch Baptist Church	West Baton Rouge	2778 Billups Ln.	Brusly	70719	30.3803462	-91.2444407	1
5	Arc of Safety	East Baton Rouge	3823 Choctaw Dr	Baton Rouge	70806	30.471788	-91.1519841	1
6	Ascension Parish Government	Ascension	615 E. Worthey Rd.	Gonzales	70737	30.2256776	-90.9162052	2
7	Assumption Parish Police Jury	Assumption	4813 Hwy-1	Napoleonville	70390	29.9416179	-91.0245497	5
8	Bayou Sorrel Bayou Pigeon Fire Station	Iberville	35575 Hwy 75	Plaquemine	70764	30.2141302	-91.0956818	2

[SD]After the drivers expressed their confusion about whether a stop on their route was a delivery or pickup, the address book was updated in Samsara (Figure 14). Using the raw agency data from the sponsor, each agency was tagged as a delivery, pickup, both, or distribution. Since tags are not visible during routes, the notation “(P)”, “(D)”, “(P&D)”, and CFSP was placed at the beginning of the name of the agency, respective to its tag, ensuring visibility during routes.

Name ▲	Address	Tags	Safety Event Exclusions	Camera Recording	Notes	Address Type	
(D) Buena Vista ID: 146164336	10633 LA-18, St James, LA 70086, USA	Delivery		Default to Vehicle Setting		Normal Geofence	Edit Remove
(D) Alsen Com... ID: 120840387	610 Old Rafe Mayer Rd, Baton Rouge, LA 70807, USA	Delivery		Default to Vehicle Setting		Normal Geofence	Edit Remove
(D) Antioch BC ID: 184087848	2778 Billups Ln, Brusly, LA 70719, USA	Delivery		Default to Vehicle Setting		Normal Geofence	Edit Remove
(D) ASSUMPTI... ID: 170029398	4813 LA-1, Napoleonville, LA 70390, USA	Delivery		Default to Vehicle Setting		Normal Geofence	Edit Remove

Figure 14: Sample of Samsara Address Book.

By going through and cross-checking Samsara’s agency information with the Excel sheet provided, many discrepancies were addressed. These differences included the spelling, the

number of times an agency was in the address book, the address itself, etc. The addresses were standardized across both platforms (Samsara and Excel) to remove discrepancies and confusion.

Samsara Implementation Guidelines

[AS]As discussed, the team's original goal was to create an optimized operational schedule that could be compared to the previous year's schedule and to calculate a tangible efficiency increase. While the team was unable to implement new changes, operational schedules were still created with the Samsara software. This section details the team's approach to using Samsara for optimization and determining delivery times. The food bank can continue Samsara implementation by following the strategy below.

The team placed each truck's stops for each day on the operational schedule in no specific order after grouping stops in MyMaps. The order was then decided by using the "Optimize Route" feature in Samsara (Figure 15 and Figure 16). Samsara uses several factors to create optimal routes besides location, including road conditions, the least fuel-intensive path, and distance. The steps to create optimal routes include setting a departure time from the Greater Baton Rouge Food Bank, adding a slot for each stop for the truck that day, selecting the agency from the list of updated names with addresses, adding the food bank as the final destination, and clicking the "Optimize by Shortest Distance" button. The interface will display the arrival and departure times for each stop, including the starting point.

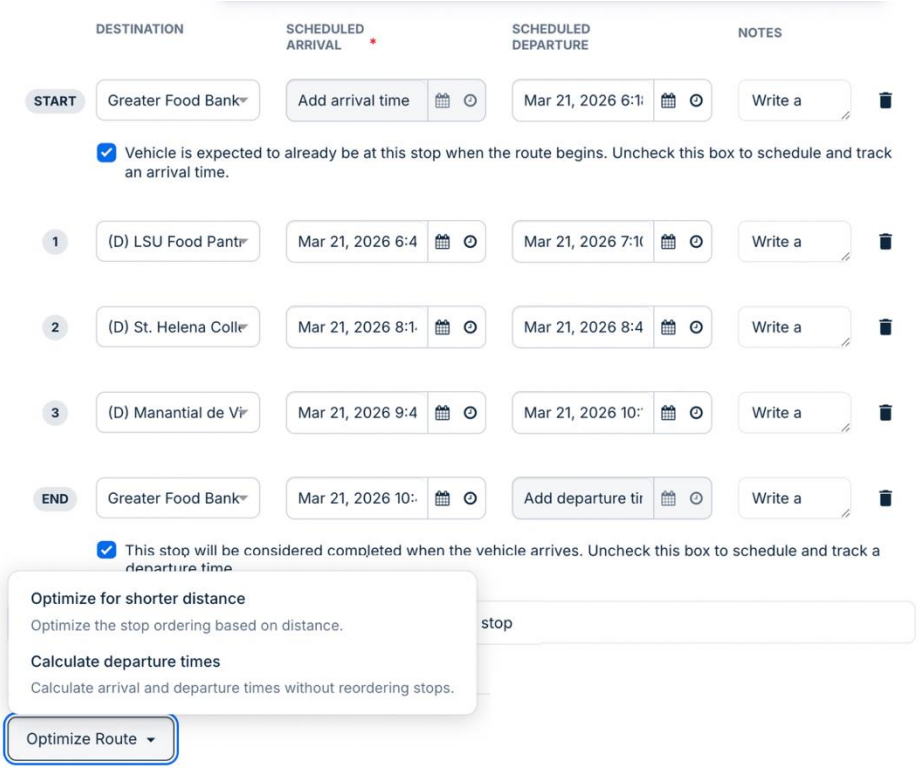


Figure 15: Samsara route creation page and Optimize Route feature.

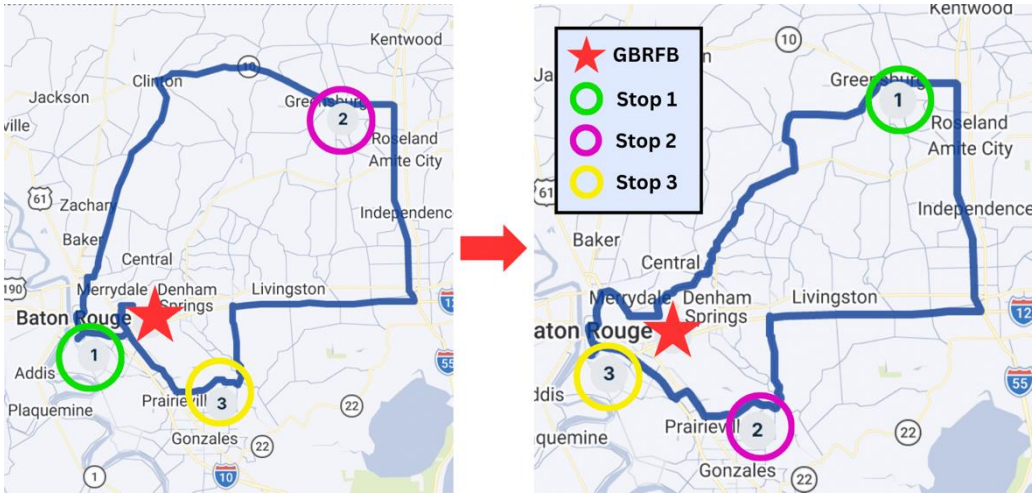


Figure 16: Before and After Samsara's "Optimize Route" Feature.

The food bank departure times range from 6:00 AM–7:30 AM. The first destination's arrival time often falls between 7:00 AM–8:00 AM, and the second between 8:00 AM–8:30 AM. After optimizing the route, the team arranged the stops accordingly on the schedule and noted the departure time from the food bank that would be required, as well as if the agency could not be found in the routing selection in Samsara.

By fully relying on Samsara for route creation, the food bank does not need to calculate manual delivery times, avoiding over- or underestimating delivery windows for partner agencies. Samsara integrates a 30-minute delivery window as well, giving exact delivery times that take unloading into account. After optimizing routes within Samsara, the food bank is able to adjust the operational calendar shown in Figure 17.

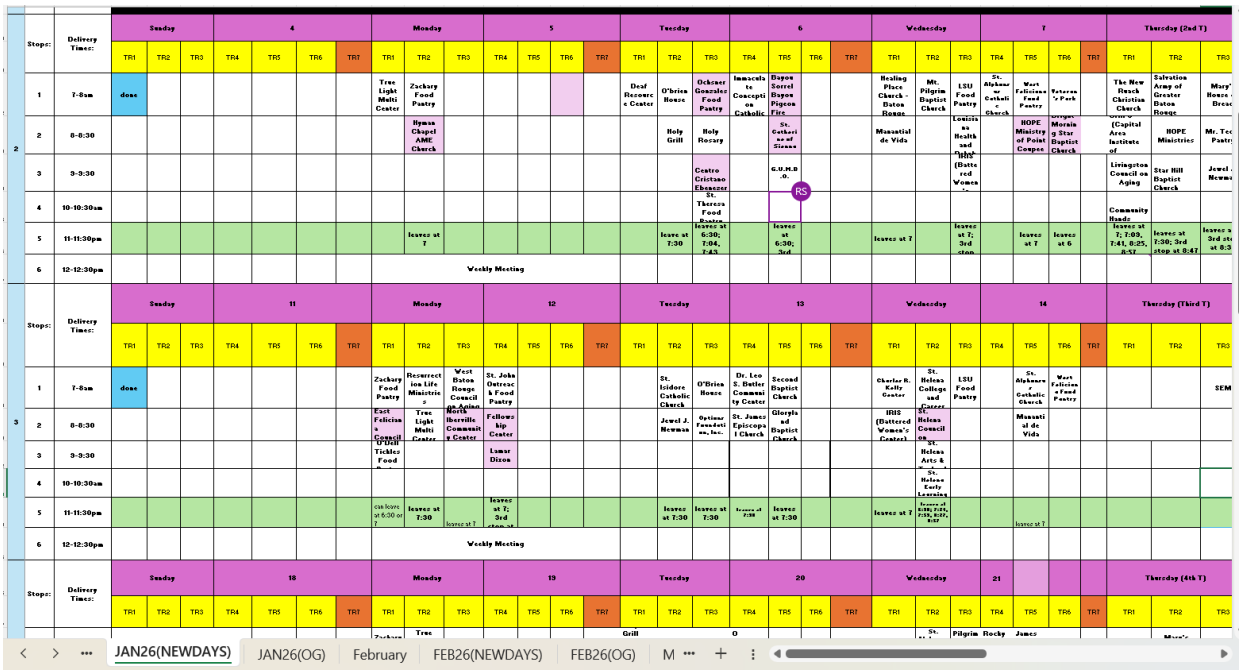


Figure 17: Sample of 2026 Operational Calendar. Notes on departure times are found in green.

This feature also helps optimize routes beyond the zoning strategy due to the program's AI-driven ability to find the most optimal routes. While zoning provides the framework to find recommended delivery days for specific agencies, the most optimal routes based on distance are needed to create the final schedule.

Progress Monitoring

[AS]A Power BI dashboard was created to allow the food bank to visualize the trucks' performance metrics. Management mentioned they wanted to eventually move to using this application, so the team developed a prototype of useful metrics they would be interested in seeing.

The first step was to connect the data to the dashboard using a link to the software. The developer tools in Samsara were not showing the necessary data in the script, which was an indicator to use an API token. The team received permission from management, who also gave full admin access to the team to complete this task. Settings had to be adjusted in Samsara to turn it on and allow certain metrics to be read by the token. From there, advanced web settings were used to import the data. Expanding the data in the table in the query allowed the team to build visuals properly. Two queries were also created for start and end dates, which tell Power BI to pull data from specific dates ranging from the first day of the current year to the current day, so that the data updates daily. Finally, graphs were built by dragging desired factors to the x and y axes. The visuals created include fuel consumption, idle time, average miles per gallon, distance travelled, and estimated fuel energy cost, as well as a filter to identify trucks individually (Figure 18).

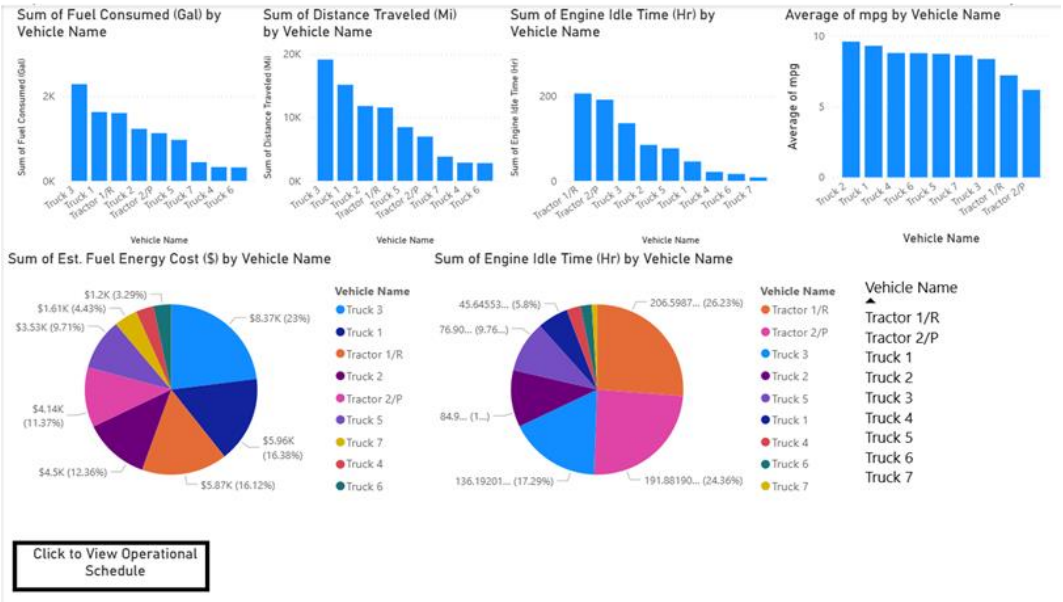


Figure 18: Power BI Dashboard.

The main benefit of the dashboard is to consolidate information that can be time-consuming to find on Samsara and to create a visual. It can all be found in one place by just opening the file. In the Samsara software, the user can see numbers on different sections of the

site, but there are no graphs to represent most of the data. The food bank would benefit greatly from being able to see the breakdown of distance travelled per truck or idle time, for example. Management can make informed decisions when assigning routes evenly if they are able to see that some trucks have higher utilization than others, so that they can be leveled. Seeing metrics broken down, side by side, such as idle time and estimated fuel cost, can also bring attention to financial issues. If management sees certain trucks leave their engines idle more often, they can address it with the drivers, for example. An SOP has been created for management to use the dashboard, replace the API token in case it expires, and create a new dashboard (SOP-004: Power BI Dashboard). The instructions are organized into 3 sections according to each topic. The SOP details the use of the filter, which isolates the metrics for the individual trucks, as well as the button that opens the operational schedule externally. The next section details replacing the API token, which can expire periodically. This entails opening the Advanced Editor, finding the API link in the script, which is highlighted in yellow in the instructions, and pasting the new one in its place. After clicking "OK" and then "Close & Apply," the dashboard should be updated with the new API link. The last section covers building a new dashboard from scratch, which will be useful in the event of the file being accidentally lost or deleted, or if management wants to create one for a different project. This last section includes obtaining the API token, filling out the "From Web" window, formatting the start date, end date, and metrics queries, copying and pasting the script into the Advanced Editor with the corresponding token, and building the visuals.

Analysis, Assessment, and Improvement

Statistical Analysis: Client Records Pivot Tool

[RS]To analyze the benefit of implementing the Client Records Pivot Tool as an aid to delivery scheduling, the team conducted a time study with a corresponding hypothesis test. The hypothesis test was conducted using a one-sample one-sided t-test, and the test is detailed below:

- H_0 (Null Hypothesis): $\mu_{tool} = \mu_{original}$
- H_1 (Alternative Hypothesis): $\mu_{tool} < \mu_{original}$

[SD] [AS] [RS]As mentioned previously, the team conducted three cycles of testing (n = 3). Meticulous notes were taken during the time frame, and the team found that food bank staff easily adapted to the tool. The employee being observed mentioned that the tool expedited the process and was easier than handling the paper binder.

	Cycle			Notes
Element	March	April	May	
Sunday	Sunday (start): 9:31AM	Sunday (start): 9:07AM	Sun: 10:30AM 6s End	March Notes: Had to assist with selecting Sunday on drop down, and then stopped to check paper calendar for 20 s (discussed a change that was recently made, but does not affect study). Mentions that in the past there were interruptions as she makes calendar. 9:44 AM and 9:46 AM someone came in to ask about an order. Monty came in at 10:01 AM for a second PAUSE 10:02 AM to 10:07 AM for an urgent order placement. Checks phone at 10:08 AM to take a call.
Monday	33s Monday: 9:31AM	22s: Tuesday: 9:07AM	Monday (start): 9:56AM	April Notes: starts typing in instead of copying and pasting, Also does Tuesday before Monday someone comes in to talk at 9:15 AM for 3 seconds stopped at 9:40 to talk until 9:43 stops at 9:52 for a few seconds to check an order
Tuesday	10 min 38s Tues: 9:41AM	13 min 36 s Mon: 9:20-9:21AM	6min 24s Tues: 10:03AM	May Notes: Starts talking at 10:15AM for about 8 seconds while typing
Wednesday	9 min 7s Wed: 9:50AM	*pause* 9:24-9:26 6min 3s	6min 15s Wed: 10:09AM	
Thursday	11 min 5s Pause Thurs: 10:07AM	10min 18s Thurs: 9:36AM	5min 20s Thurs: 10:15AM	
Friday	10 min 20s Fri: 10:17AM	9min 24s Fri: 9:45AM	7 min 15 s Fri: 10:22AM	
Saturday	9 min 24s Sat: 10:26 AM	7min 9s Sat: 9:52AM	4 min 41 s Sat: 10:27	
TOTAL	3 min 14s Done **53.8 min**	2min 33 s Done at 9:55 **49 min 25s**	2 min 30s (Go to Sunday) **32 min 31s**	
Feedback from Cherelle:	quicker than going through paper binders			
7 minutes per agency orginally. With 116 agencies, original time to produce a calendar was ~67 minutes				

Figure 19: Time Study Datasheet.

The time study datasheet is visualized in Figure 19. Due to time constraints, the team requested an estimated time for schedule creation to be used as the baseline for comparison. The employee in charge of producing the calendars estimated ~67 minutes to complete one month of deliveries (with 116 agencies), which was based on the time it takes to complete a single agency's request over the span of a year.

After completing the study, it was found that the total times to complete March, April, and May calendars were 53.8 minutes, 49.25 minutes, and 32.5 minutes, respectively. Averaging cycle times resulted in a value of 45.18 minutes required per month for schedule creation with the tool. Using this value and the following formula, the test statistic was determined to be -3.37 at a level of 95% confidence.

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

Using Excel, the critical statistic was found to be -2.92, and the p-value for this experiment was found to be 0.039. Calculations can be found in the Appendix (Figure 23).

$$-3.37 < -2.92$$

$$0.039 < 0.05$$

The test statistic was less than the critical value, and the p-value was also less than the significance value of 0.05, indicating that the null hypothesis was rejected, and the schedule creation time was shortened through the use of the Client Records Pivot Tool.

Statistical Analysis: Samsara Optimal Route Creation

[RS]The food bank currently uses Samsara to store route information and record key driver metrics, and this program includes an AI-based route organization tool. This tool claims to

organize routes by the shortest possible distance, and the team wanted to confirm this as fact, so the food bank can fully rely on Samsara for all route generation.

A Non-Inferiority paired t-test was used to perform the hypothesis test, and the experimental setup is explained below:

- $D = Miles_{Samsara} - \min[Miles_{Optimal}]$
- $M_{NI} = 2\ miles$
- $H_0(Null\ Hypothesis)$: $\mu_0 \geq 2\ miles$
- $H_1(Alternative\ Hypothesis)$: $\mu_D < 2\ miles$

Non-Inferiority testing is prominent in medical fields, as it is used to determine if a new medicine is statistically equal to or better than the current treatment (Hahn, 2012). This type of testing is beneficial in this study, as it sets the framework to determine if Samsara's routes are equivalent to the quickest possible route. The margin of Non-Inferiority ($M_{NI} = 2$) was set at 2 miles, meaning that Samsara's results were deemed equivalent if they were within the two-mile range of the global optimum. The value of 2 miles was selected as it represents ~5% of the average route distance, which the team calculated based on total routes from the analyzed January, February, and March data.

[MP]To collect data for the test, the month of March was analyzed, and all routes with multiple stops were identified (Appendix Figure 24). These routes were enumerated to find every possible route combination, so that the shortest route could be identified. For example, a route with 3 stops could have 9 (3!) possible route combinations. Google Maps was used as the distance engine to calculate mileage, and all possible mileages were recorded, as well as mileages corresponding to Samsara's recommended routes. The data sheet is visualized in Figure 20. The full data set can be found in the Appendix (Figure 24).

ORIGINAL					SAMSARA					
Agency	Date	Truck	Possible Mileage		Agency	Date	Truck	MILES	Difference	truck 1
True Light Multi Center	Monday, March 2	1	13.4,13.8		Interfaith Federation (Holy Grill)	Monday, March 2	1	13.4	0	truck 2
Holy Grill (Interfaith Federation)	Monday, March 2	1			True Light Multi Center	Monday, March 2	1		0	truck 3

Figure 20: Mileage Data Sheet for Non-Inferiority Testing.

[RS]The shortest possible route from all enumerated routes served as the optimum. After comparing both sets of data, Samsara routes were ~0.35 miles longer than the optimal route (D). Using Excel, a sample size of 40 (n = 40), and the following formula, the test statistic for the paired t-test was found to be -5.47.

$$t = \frac{\overline{D} - M_{NI}}{s/\sqrt{n}}$$

With an alpha value of 0.05 and a degree of freedom of 39, the critical test statistic was then found to be -1.685. The p-value was also calculated through Excel, and it was found to be a very small value under the significance level of 0.05 ($p = 1.6 \times 10^{-6}$). All statistical results can be found in the Appendix (Figure 23).

$$-5.47 < -1.685$$

The test value was less than the critical value, which led the team to reject the null hypothesis and confirm that Samsara's optimized routes were equivalent alternatives to optimal routes, within 2 miles. While this test is significant and is credible in proving Samsara's capability, it is quasi-statistical in nature. The specific days and routes were chosen at random, but the routes themselves are concrete in their total distance and cannot be changed. However, this test confirms that the food bank can rely on Samsara's AI optimization to produce optimal routes in record time, and by using this application fully, the food bank can reduce total mileage over time.

Project Continuation

Standard Operating Procedures have been given to management to support the continuation of this project beyond the team's involvement. The SOPs provide detailed step-by-step instructions on how to use each of the recommendations, including the Client Records Pivot Tool, Power BI dashboard, driver suggestions, and Google Form. Management was given the opportunity to provide feedback on each SOP's clarity and usability in the Management Feedback Form. The team also walked through demonstrations of all the features to ensure understanding of the application beyond the SOP. All SOPs are maintained in Google Docs, so they will be stored securely and remain editable. This ensures that the improvements made are sustainable long-term and that management can adapt documents to fit their needs. All SOPs can be found at the end of the document's Appendix.

Feedback Collection

[RS]In the spring semester, management was provided with a feedback form (Figure 21). Each member of management (including the corporate sponsor, the transportation team, and those in agency relations) will give their input on the team's suggested system. This data will help the team to continue to personalize this delivery organization system to meet the needs of those involved. The team can evaluate the success of proposed recommendations by surveying the individuals heavily involved in the process.

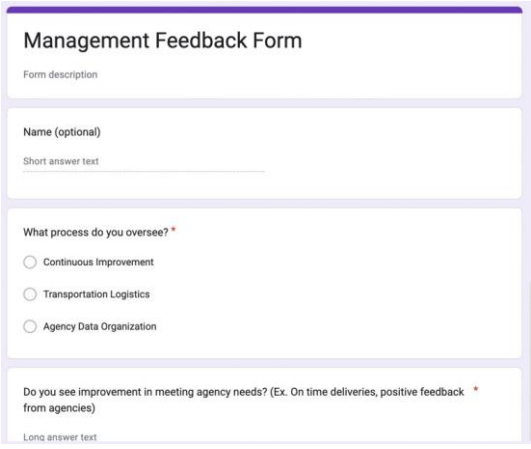


Figure 21: Management Feedback Form.

The four feedback-based questions asked in the form are as follows:

1. Do you feel as though the delivery optimization strategy that the team has presented will aid in reducing metrics in the upcoming years? Why do you feel this way?
2. Are the tools presented so far (Pivot Tool for agencies, SOPs, zones, Power BI) easy to understand, and are they applicable to your work? Are there any improvements in this area that you would like to see?
3. What are your suggestions for further improvement by the LSU team?
4. Is there any positive feedback you would like to share?

All three key stakeholders from the food bank left very positive reviews using the feedback form. All individuals agreed that the strategy will reduce key metrics (mileage and fuel usage) in the upcoming years. One stakeholder has verbally mentioned that the projected savings of an estimated 20% mileage reduction per month, while using the clustering approach, inspire the team to continue with this project (Table 9). The responders also stated that the tools are applicable for their work, especially the Client Records Pivot Tool and the implementation of Google Forms for future digital data collection. All SOPs were rated as easy to understand and implement. One suggestion for further improvement was for the Power BI portion of the project, but in regard to the food bank team not having the software installed yet. However, they are working towards buying the license so that they can use the team's provided dashboard in the near future. The food bank stakeholders also mentioned that more practice with the tools will benefit the team in terms of efficiency in the upcoming years.

Each of the above answers to the questions can be found in the Appendix. Management was overall satisfied with the work provided by the team (Figure 26, Figure 27, Figure 28, Figure 29).

Economic and Environmental Justification

[RS] [SD] An analysis of mileage was performed on the routes before and after rearranging the delivery schedule. An Excel sheet was created, listing the name of the stop, addresses pulled from Samsara, truck number, day, day of the week, and mile column. Using the operational schedule created from the requests and Samsara's "Optimize Route" feature, the order of deliveries was pasted into Google Maps, and the total mileage for the route for each truck, on each day, is recorded on the right. The task of collecting mileage data was divided by month. The colors correspond to the truck numbers (Figure 22).

Stop	Address	Truck	Day	Day of We	
True Light Multi Center	3836 North Street, Baton Rouge, LA	Truck 1	Feb 2nd	Monday	12.7
Zachary Food Pantry	5640-A Rollins Rd, Zachary, LA, 707	Truck 2	Feb 2nd	Monday	36.7
Immaculate Conception Cath	1565 Curtis St. Baton Rouge, LA 70	Truck 1	Feb 3rd	Tuesday	17.6
O'Brien House	446 North 12th St, Baton Rouge, LA	Truck 2	Feb 3rd	Tuesday	16.2
Holy Grill (Interfaith Federatio	3112 Convention St/6117 Cadillac	Truck 2	Feb 3rd	Tuesday	
Deaf Resource Center	16288 Tiger bend RD	Truck 3	Feb 3rd	Tuesday	18.2
G.U.M.B.O (Global United Miss	23440 Railroad Ave, Plaquemine, LA	Truck 4	Feb 3rd	Tuesday	89.3
St. Catherine of Sienna	421 St. Patrick Street, Donaldsonvil	Truck 4	Feb 3rd	Tuesday	
St. Theresa Food Pantry	1105 N Burnside Ave, Gonzales, LA	Truck 5	Feb 3rd	Tuesday	55.3
Holy Rosary Roman Catholic (	44450 LA-429, St Amant, LA 70774	Truck 5	Feb 3rd	Tuesday	
Hyman Chapel AME Church	10732 Reily St, Clinton, LA 70722	Truck 6	Feb 3rd	Tuesday	69.1

Figure 22: Sample of Excel Route Mileage Sheet

The summation of mileage for each week in January and February was calculated and shown in the table below. The difference in miles scheduled between the original and rearranged routings was calculated as a percentage change, with the negative values indicating a decrease, therefore saving fuel (Table 9).

Table 9: Mileage Evaluation

	January			Feburary		
Week	**Original (miles)**	**Rearranged (miles)**	**% Change**	**Original (miles)**	**Rearranged (miles)**	**% Change**
Week 1	1338.9	1074.5	-19.75%	753.3	601.1	-20.20%
Week 2	1077.4	750.5	-30.34%	1358.4	1101.4	-18.92%
Week 3	1209.8	909.4	-24.83%	540	430	-20.37%
Week 4	371.9	364.4	-2.02%	1143.1	948.1	-17.06%
Total (miles)	3998	3098.8	-22.49%	3794.8	3080.6	-18.82%
Total (gallons)	484	375.2	-22.49%	459.4	373	-18.82%

By implementing the rearranged schedule, the GBRFB can save 18% - 22% of fuel costs, due to the reduction in total miles traveled. Further economic evaluation, including actual diesel prices and consumption per truck, is outside of the scope of this project. However, by implementing this optimization strategy, the food bank will have monthly savings much greater than the original goal of 5% - 10% savings.

By evaluating the mileage before and after visual clustering, the potential savings for the year can be extrapolated. Combined, this strategy led to a decrease of 195.2 gallons of diesel gas between January and February. On average, that is ~98 gallons a month, leading to 1176 gallons a year. Although gas prices fluctuate due to socioeconomic and political conditions, a value of $3.50/gallon is used to conclude that there is a potential savings of $4116/year.

Environmental impact can also be measured through the Environmental Protection Agency's standard conversion factors. One gallon of diesel gas equates to ~10.2k grams of CO_2 emissions (US EPA, 2025). The total sum of gallons saved through implementation of the rearranged schedule, for January and February, is 195.2 gallons. By multiplying gallons saved by the conversion factor, the total metric tons of carbon dioxide avoided can be calculated. By reducing gas consumption, the GBRFB is working towards reducing the carbon footprint in Louisiana. If the suggested alternative is implemented, the food bank would reduce emissions by about 2 metric tons of CO_2 in January and February alone.

$$195.2\ gallons * 10.18 * 10^{-3}\ metric\ tons \frac{CO_2}{gallon} = \sim 2\ metric\ tons\ of\ CO^2$$

One main aspect of this project was to ensure that drivers had similar workloads when it came to the delivery schedule. After calculating the mileage for January and February, the mileage per truck was summarized. The team made some slight adjustments within the operational schedules for the original calendar and the rearranged calendar to ensure equal distribution, and then it was confirmed that drivers had similar route complexity and lengths. All drivers were assigned 14%-19% of the total miles each month, which will lead to standardized utilization throughout the year (Table 10 and Table 11). The food bank currently does not have a

process for ensuring equal routes per driver. However, the team can conclude that by calculating mileage after calendar creation, miles can be monitored, and simple switches of route assignments can standardize the drivers' assignments when needed.

Table 10: Truck Utilization for January

January		
Truck	Requested: % of Total Miles	Optimized: % of Total Miles
Truck 1	17%	18%
Truck 2	17%	18%
Truck 3	17%	18%
Truck 4	16%	15%
Truck 5	15%	16%
Truck 6	18%	15%

Table 11: Truck Utilization for February

February		
Truck	Requested: % of Total Miles	Optimized: % of Total Miles
Truck 1	16%	18%
Truck 2	19%	19%
Truck 3	17%	17%
Truck 4	16%	14%
Truck 5	17%	17%
Truck 6	15%	14%

It is essential to note that all drivers complete their routes within their 8-hour working window, with a majority of routes being far less than 8 hours. Time spent outside of deliveries is allocated to pick up orders, meaning that the drivers all have equal assignments throughout their work week.

In terms of time and employee hours saved when implementing recommendations, the team calculated the total savings of implementing the Pivot Tool and EDL. Using these tools, the employee can minimize the amount of time spent on creating the operational calendar and manually sorting through paper requests. While the wage of the employee is unknown, it can be concluded that there is substantial time savings: from 67 minutes per month to 4~5, this ~20 minute savings per month leads to 240 minutes or 4 hours saved, which can be reallocated to completing other tasks.

Conclusion

The Greater Baton Rouge Food Bank will benefit from the provided optimization strategy for many years to come. As the food bank team works to integrate optimization plans with the partner agencies, agencies can adapt their volunteer schedules to allow for fuel-efficient deliveries. With full Samsara implementation, the food bank will have clarity as to the performance of drivers and can work to be as fuel-efficient as possible. All funds saved by the food bank through optimization allow for more resources to be invested into the Greater Baton Rouge community, ensuring that "Together, we're fighting hunger."

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Appendix

Statistical analysis not pictured above is pictured below, beginning with the statistical values calculated with the time study (Figure 23).

Sample mean	45.18333333
Original time	67
SD	11.21721148
n	3
Test stat	-3.368713801
Critical stat	-2.91998558
p	0.038977378

Figure 23: Statistical analysis calculated in Excel.

The following information details the statistical analysis for Non-Inferiority testing within Excel. Each color signifies a truck assignment (Figure 24 and Figure 25).

ORIGINAL					SAMSARA				
Agency	Date	Truck	Possible Mileage	Minimum Miles	Agency	Date	Truck	Mileage	Difference
True Light Multi Center	Monday, March 2	1	13.4,13.8	13.4	Interfaith Federation (Holy Grill)	Monday, March 2	1	13.4	0
Holy Grill (Interfaith Federation)	Monday, March 2	1			True Light Multi Center	Monday, March 2	1		0
O'Brien House	Tues, March 3	3	17.9, 17.7	17.7	O'Brien House	Tues, March 3	3	17.9	0.2
St. Agnes	Tues, March 3	3			St. Agnes	Tues, March 3	3		0
Holy Rosary Roman Catholic Ch	Tues, March 3	4	55.6, 53.6	53.6	St. Theresa Food Pantry	Tues, March 3	4	55.6	2
St. Theresa Food Pantry	Tues, March 3	4			Holy Rosary Roman Catholic Churcl	Tues, March 3	4		0
Frank Sotile Pavillion	Tues, March 3	5	89.7, 90.7, 128	89.6	G.U.M.B.O (Global United Mission B	Tues, March 3	5	89.7	0.1
G.U.M.B.O (Global United Missic	Tues, March 3	5	90.5, 128, 89.6		St. Catherine of Sienna	Tues, March 3	5		0
St. Catherine of Sienna	Tues, March 3	5			Frank Sotile Pavillion	Tues, March 3	5		0
Jewel J. Newman Community Ce	Thurs, March 5	1	27.3, 27.3	27.3	Mr. Ted's Pantry (formerly Baker Pres	Thurs, March 5	1	27.3	0
Mr. Ted's Pantry (formerly Baker	Thurs, March 5	1			Jewel J. Newman Community Cente	Thurs, March 5	1		0
Mary's House of Bread	Thurs, March 5	3	19.4, 19.8	19.4	Mary's House of Bread	Thurs, March 5	3	19.4	0
Hispanic Apostolate	Thurs, March 5	3			Hispanic Apostolate	Thurs, March 5	3		0
Salvation Army of Greater Baton	Thurs, March 5	5	10.7, 11.2	10.7	Salvation Army of Greater Baton Rou	Thurs, March 5	5	10.7	0
Star Hill Baptist Church	Thurs, March 5	5			Star Hill Baptist Church	Thurs, March 5	5		0
CAIFS (Capital Area Institute of F	Thurs, March 5	6	15.3,15.9	15.3	Community Hands		6	15.3	0
Community Hands	Thurs, March 6	6			CAIFS (Capital Area Institute of Family Services)		6		0
St. Vincent De Paul / Bishop Ott	Fri, March 6	1	16, 18.7, 15.9	16	St. Vincent De Paul / Bishop Ott	Fri, March 6	1	16	0
St. Anthony Padua	Fri, March 6	1	18.8, 16.3, 16.4		Louisiana Health and Rehab (Reality	Fri, March 6	1		0
Louisiana Health and Rehab (Re	Fri, March 6	1			St. Anthony Padua	Fri, March 6	1		0
Centro Cristano Ebenezer	Fri, March 6	3	55.7, 50.2	50.2	Richard Brown 1st Pilgrim	Fri, March 6	3	55.7	5.5
Richard Brown 1st Pilgrim	Fri, March 6	3			Centro Cristano Ebenezer	Fri, March 6	3		0
Resurrection Life Ministries (forr	Mon, March 9	4	30.4, 30	30	Resurrection Life Ministries (former	Mon, March 9	4	30.4	0.4
West Baton Rouge Council on Ag	Mon, March 9	4			West Baton Rouge Council on Aging	Mon, March 9	4		0
O'Brien House	Tues, March 10	2	22, 22.6, 20	20	Dr. Leo S. Butler Community Center	Tues, March 10	2	22	2
St. James Episcopal Church	Tues, March 10	2	22.7, 20.2, 22.3		St. James Episcopal Church	Tues, March 10	2		0
Dr. Leo S. Butler Community Cer	Tues, March 10	2			O'Brien House	Tues, March 10	2		0
Gloryland Baptist Church	Tues, March 10	3	19.7, 21.3,21.7	19.5	options	Tues, March 10	3	19.7	0.2
Second Baptist Church	Tues, March 10	3	21.4, 22.4, 19.5		Second Baptist Church	Tues, March 10	3		0
options	Tues, March 10	3			Gloryland Baptist Church	Tues, March 10	3		0
Jewel J. Newman Community Ce	Tues, March 10	4	23.4, 23.8	23.4	St. Isidore Catholic Church	Tues, March 10	4	23.4	0
St. Isidore Catholic Church	Tues, March 10	4			Jewel J. Newman Community Cente	Tues, March 10	4		0
St. Alphonsus Catholic Church	Wed, March 11	3	16.3, 15.9	15.9	IRIS (Battered Women's Center)	Wed, March 11	3	16.3	0.4
IRIS (Battered Women's Center)	Wed, March 11	3			St. Alphonsus Catholic Church	Wed, March 11	3		0
Healing Place Church - Baton Rc	Wed, March 11	4	34.9, 34.9	34.9	Healing Place Church - Baton Rouge	Wed, March 11	4	34.9	0
Manantial de Vida	Wed, March 11	4			Manantial de Vida	Wed, March 11	4		0

Veteran's Park	Wed, March 11	5	120, 120	120	Lamar Dixon	Wed, March 11	5	120	0
Lamar Dixon	Wed, March 11	5			Veteran's Park	Wed, March 11	5		0
St. Helena Arts & Technology Acz	Wed, March 11	6	81.1, 81.5, 83.9	81.1	St. Helena College and Career Acad	Wed, March 11	6	81.1	0
St. Helena College and Career A	Wed, March 11	6	81.5, 83.9, 81.1		St. Helena Early Learning Center	Wed, March 11	6		0
St. Helena Early Learning Center	Wed, March 11	6			St. Helena Arts & Technology Acader	Wed, March 11	6		0
North Iberville Community Cent	Thurs, March 12	3	52.8, 53.1	52.8	North Iberville Community Center	Thurs, March 12	3	52.8	0
Douglas Avenue Baptist Church	Thurs, March 12	3			Douglas Avenue Baptist Church	Thurs, March 12	3		0
Redemption Rose Fellowship Mi	Thurs, March 12	4	40.7, 40.7	40.7	Redemption Rose Fellowship Minist	Thurs, March 12	4	40.7	0
Livingston Council on Aging	Thurs, March 12	4			Livingston Council on Aging	Thurs, March 12	4		0
St. Helena Council on Aging	Thurs, March 12	5	85.7,84.6	84.6	St. Helena Council on Aging	Thurs, March 12	5	85.7	1.1
Rocky Hill A.M.E. Church	Thurs, March 12	5			Rocky Hill A.M.E. Church	Thurs, March 12	5		0
St. Vincent De Paul / Bishop Ott	Fri, March 13	1	16.9, 16.9	16.9	Scott-Gilchrist Bethel AME Church-	Fri, March 13	1	16.9	0
Scott-Gilchrist Bethel AME Chur	Fri, March 13	1			St. Vincent De Paul / Bishop Ott	Fri, March 13	1		0
Bayou Sorrel Bayou Pigeon Fire S	Fri, March 13	6	158, 157, 141	141	Bayou Sorrel Bayou Pigeon Fire Stat	Fri, March 13	6	158	17
Bright Morning Star Baptist Chur	Fri, March 13	6	158, 157, 141		Bright Morning Star Baptist Church	Fri, March 13	6		0
Word Teaching Ministries	Fri, March 13	6			Word Teaching Ministries	Fri, March 13	6		0
Albany First Baptist Church	Sun, March 15	1	72, 72, 72.9	71.1	Albany First Baptist Church	Sun, March 15	1	72	0.9
Faith Crossing UMC	Sun, March 15	1	71.1, 71.5, 96.1		Faith Crossing UMC	Sun, March 15	1		0
Holden Baptist Church	Sun, March 15	1			Holden Baptist Church	Sun, March 15	1		0
Holy Grill (Interfaith Federation)	Mon, March 16	3	29.8, 29.7	29.7	West Baton Rouge Council on Aging	Mon, March 16	3	29.8	0.1
West Baton Rouge Council on Ag	Mon, March 16	3			Holy Grill (Interfaith Federation)	Mon, March 16	3		0
Zachary Food Pantry	Mon, March 16	5	73.5, 73.9	73.5	Zachary Food Pantry	Mon, March 16	5	73.5	0
Hyman Chapel AME Church	Mon, March 16	5			Hyman Chapel AME Church	Mon, March 16	5		0
O'Brien House	Tues, March 17	2	32.7, 33	32.7	O'Brien House	Tues, March 17	2	32.7	0
Alsen St. Irma Lee Community Vi	Tues, March 17	2			Alsen St. Irma Lee Community Villag	Tues, March 17	2		0
Broadmoor High School "Buccar	Tues, March 17	4	20.5, 20.5	20.5	Deaf Resource Center	Tues, March 17	4	20.5	0
Deaf Resource Center	Tues, March 17	4			Broadmoor High School "Buccaneer	Tues, March 17	4		0
Mt. Pilgrim Baptist Church	Wed, March 18	2	65.1, 64.5	64.5	West Feliciana Food Pantry	Wed, March 18	2	65.1	0.6
West Feliciana Food Pantry	Wed, March 18	2			Mt. Pilgrim Baptist Church	Wed, March 18	2		0
Jewel J. Newman Community Ce	Thurs, March 19	3	28.8, 28, 29.2	28	Camphor Memorial	Thurs, March 19	3	28.8	0.8
Mr. Ted's Pantry (formerly Baker	Thurs, March 19	3	30.3, 31.3, 29.7		Jewel J. Newman Community Cente	Thurs, March 19	3		0
Camphor Memorial	Thurs, March 19	3			Mr. Ted's Pantry (formerly Baker Pre	Thurs, March 19	3		0
Salvation Army of Greater Baton	Thurs, March 19	4	14, 14.1	14	Salvation Army of Greater Baton Rou	Thurs, March 19	4	14	0
Hispanic Apostolate	Thurs, March 19	4			Hispanic Apostolate	Thurs, March 19	4		0
CAIFS (Capital Area Institute of F	Thurs, March 19	6	15.3, 15.9	15.3	Community Hands	Thurs, March 19	6	15.3	0
Community Hands	Thurs, March 19	6			CAIFS (Capital Area Institute of Fam	Thurs, March 19	6		0
Emmanuel Baptist Church	Sat, March 22	1	96.1, 96.1	96.1	Emmanuel Baptist Church	Sat, March 22	1	96.1	0
Le Garde Manger de Paincourtvil	Sat, March 22	1			Le Garde Manger de Paincourtville	Sat, March 22	1		0
Second Baptist Church	Tues, March 24	3	17.3, 17.2, 20	17.2	Maison Des Ami	Tues, March 24	3	17.3	0.1
O'Brien House	Tues, March 24	3	20, 17.4, 17.2		O'Brien House	Tues, March 24	3		0
Maison Des Ami	Tues, March 24	3			Second Baptist Church	Tues, March 24	3		0
Unity Church of God in Christ	Fri, March 27	1	38.1,38.4	38.1	Women's Community Rehab Center	Fri, March 27	1	38.1	0
Women's Community Rehab Cer	Fri, March 27	1			Unity Church of God in Christ	Fri, March 27	1		0
OMG (Outstanding Mature Girlz	Fri, March 27	2	22.4, 22.9	22.4	The Store @ BRCC	Fri, March 27	2	22.4	0
The Store @ BRCC	Fri, March 27	2			OMG (Outstanding Mature Girlz Org	Fri, March 27	2		0
Scott-Gilchrist Bethel AME Chur	Fri, March 27	3	16.9, 16.9	16.9	Scott-Gilchrist Bethel AME Church-	Fri, March 27	3	16.9	0
St. Vincent De Paul / Bishop Ott	Fri, March 27	3			St. Vincent De Paul / Bishop Ott	Fri, March 27	3		0
C.E.O Food Pantry (Mt. Calvary)	Fri, March 27	4	103, 103	103	C.E.O Food Pantry (Mt. Calvary)	Fri, March 27	4	103	0
St. Paul Community Agency	Fri, March 27	4			St. Paul Community Agency	Fri, March 27	4		0
Carl F. Grant Civic Center	Fri, March 27	6	55.6, 55.6	55.6	Carl F. Grant Civic Center	Fri, March 27	6	55.6	0
United Community Food Pantry	Fri, March 27	6			United Community Food Pantry	Fri, March 27	6		0
O'Brien House	Tues, March 31	4	32.5, 32.8	32.5	O'Brien House	Tues, March 31	4	32.5	0
Mr. Ted's Pantry (formerly Baker	Tues, March 32	4			Mr. Ted's Pantry (formerly Baker Pre	Tues, March 32	4		0

Figure 24: Data sheet for Non-Inferiority Testing. Each color represents a truck assignment.

Sample mean	0.35
SD	1.905797965
n	40
Test statistic	-5.466345785
Critical stat	
p	1.42516E-06

Figure 25: Statistical values found in Excel for Non-Inferiority testing.

Do you feel as though the delivery optimization strategy that the team has presented will aid in reducing metrics in the upcoming years? Why do you feel this way?

3 responses

Yes, I believe the delivery optimization strategy will help reduce key metrics in the coming years. By implementing the optimized plan, agencies within the same area will be scheduled on the same delivery day, rather than selecting different days individually. This approach will reduce travel time, improve route efficiency, and lower fuel consumption, which leading to better overall performance and cost savings.

Yes. I believe it will help our company's operation in the future.

Yes, Both the digital scheduler and the clustering should help

Figure 26: Question 1 of Management Feedback Form with responses.

Are the tools presented so far (pivot tool for agencies, SOP's, zones, PowerBI) easy to understand, and are they applicable to your work? Are there any improvements in this area that you would like to see?

3 responses

Yes, the pivot tool for agencies is especially helpful, and I use it on a regular basis to support my day-to-day tasks with order placing without having to carry a binder around.

I would like more opportunities to work with Power BI, as I believe gaining deeper knowledge and hands-on experience with it would further improve efficiency and reporting capabilities.

Yes. Everything is great.

Yes they were easy to understand and read. Sop's will be used in our next iteration of driver SOP's

Figure 27: Question 2 of Management Feedback Form with responses.

What are your suggestions for further improvement by the LSU team?

3 responses

Honestly, the team did an exceptional job. Even coming in with a set plan and having to pivot and adjust along the way, everything was handled smoothly and without showing any pressure. That level of adaptability was impressive.

My only suggestion would be to continue building on what's already working well—possibly by incorporating more hands-on training or real-time application of the tools to further strengthen understanding and consistency.

No suggestions. This team did a fantastic job.

I think yall did great taking on a very ambitious project.

Figure 28: Question 3 of Management Feedback Form with responses.

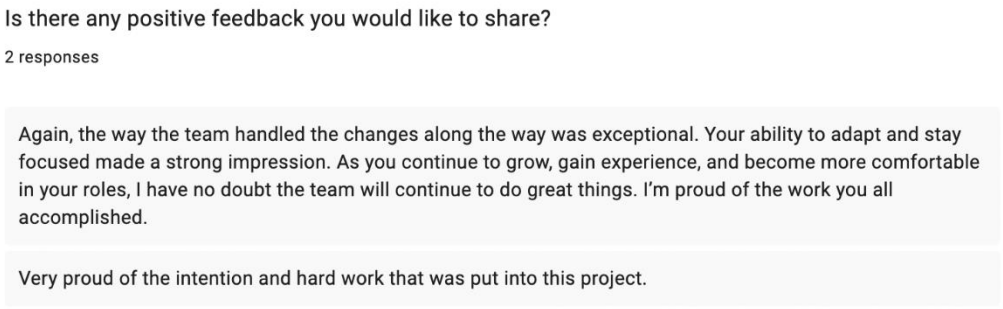


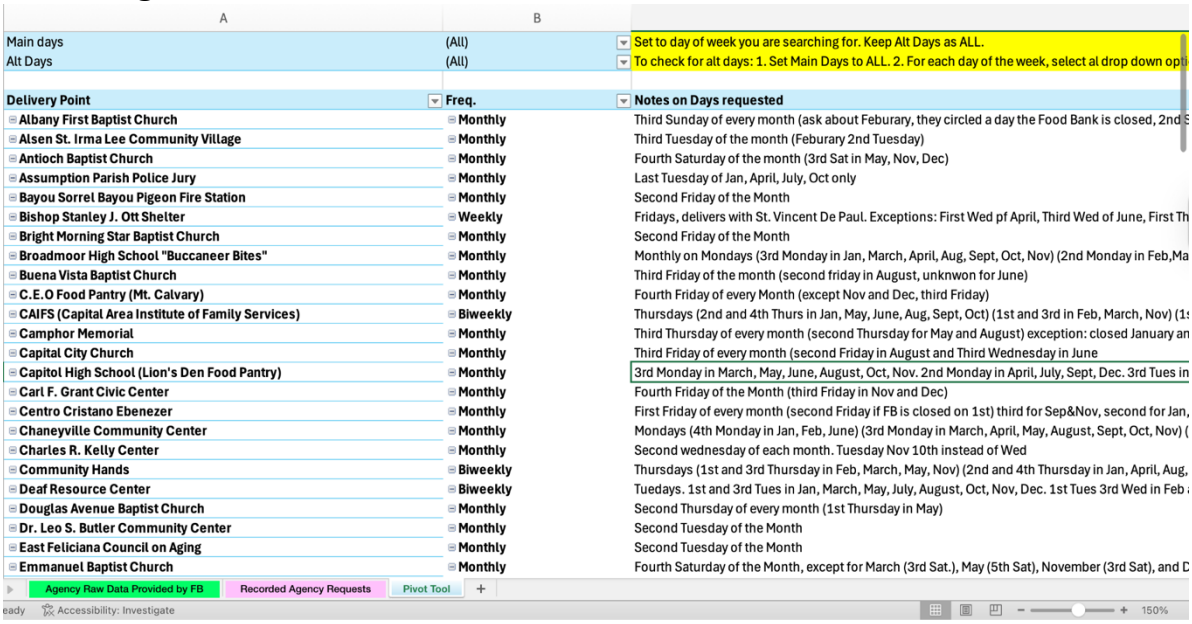
Figure 29: Question 4 of Management Feedback Form with responses.

Standard Operating Procedures

SOP-001: Client Record Pivot Tool

[RS]The purpose of this Standard Operating Procedure is to instruct the food bank on how to use and recreate the Client Records Pivot Tool. Part 1 of the SOP details how to operate the tool, while Part 2 explains how to recreate the tool year to year. The tool itself has been created in Excel, but all data collection occurs in the Google Suite. The tool explains how to transfer information to make the easiest-to-operate tool.

Part 1: Using the Client Records Pivot Tool



	A	B	
	Main days	(All)	Set to day of week you are searching for. Keep Alt Days as ALL.
	Alt Days	(All)	To check for alt days: 1. Set Main Days to ALL. 2. For each day of the week, select al drop down op
	Delivery Point	Freq.	Notes on Days requested
	Albany First Baptist Church	Monthly	Third Sunday of every month (ask about Feburary, they circled a day the Food Bank is closed, 2nd S
	Alsen St. Irma Lee Community Village	Monthly	Third Tuesday of the month (Feburary 2nd Tuesday)
	Antioch Baptist Church	Monthly	Fourth Saturday of the month (3rd Sat in May, Nov, Dec)
	Assumption Parish Police Jury	Monthly	Last Tuesday of Jan, April, July, Oct only
	Bayou Sorrel Bayou Pigeon Fire Station	Monthly	Second Friday of the Month
	Bishop Stanley J. Ott Shelter	Weekly	Fridays, delivers with St. Vincent De Paul. Exceptions: First Wed pf April, Third Wed of June, First Th
	Bright Morning Star Baptist Church	Monthly	Second Friday of the Month
	Broadmoor High School "Buccaneer Bites"	Monthly	Monthly on Mondays (3rd Monday in Jan, March, April, Aug, Sept, Oct, Nov) (2nd Monday in Feb,Ma
	Buena Vista Baptist Church	Monthly	Third Friday of the month (second friday in August, unknwon for June)
	C.E.O Food Pantry (Mt. Calvary)	Monthly	Fourth Friday of every Month (except Nov and Dec, third Friday)
	CAIFS (Capital Area Institute of Family Services)	Biweekly	Thursdays (2nd and 4th Thurs in Jan, May, June, Aug, Sept, Oct) (1st and 3rd in Feb, March, Nov) (1
	Camphor Memorial	Monthly	Third Thursday of every month (second Thursday for May and August) exception: closed January an
	Capital City Church	Monthly	Third Friday of every month (second Friday in August and Third Wednesday in June
	Capitol High School (Lion's Den Food Pantry)	Monthly	3rd Monday in March, May, June, August, Oct, Nov. 2nd Monday in April, July, Sept, Dec. 3rd Tues in
	Carl F. Grant Civic Center	Monthly	Fourth Friday of the Month (third Friday in Nov and Dec)
	Centro Cristano Ebenezer	Monthly	First Friday of every month (second Friday if FB is closed on 1st) third for Sep&Nov, second for Jan,
	Chaneyville Community Center	Monthly	Mondays (4th Monday in Jan, Feb, June) (3rd Monday in March, April, May, August, Sept, Oct, Nov) (
	Charles R. Kelly Center	Monthly	Second wednesday of each month. Tuesday Nov 10th instead of Wed
	Community Hands	Biweekly	Thursdays (1st and 3rd Thursday in Feb, March, May, Nov) (2nd and 4th Thursday in Jan, April, Aug,
	Deaf Resource Center	Biweekly	Tuedays. 1st and 3rd Tues in Jan, March, May, July, August, Oct, Nov, Dec. 1st Tues 3rd Wed in Feb
	Douglas Avenue Baptist Church	Monthly	Second Thursday of every month (1st Thursday in May)
	Dr. Leo S. Butler Community Center	Monthly	Second Tuesday of the Month
	East Feliciana Council on Aging	Monthly	Second Tuesday of the Month
	Emmanuel Baptist Church	Monthly	Fourth Saturday of the Month, except for March (3rd Sat.), May (5th Sat), November (3rd Sat), and D

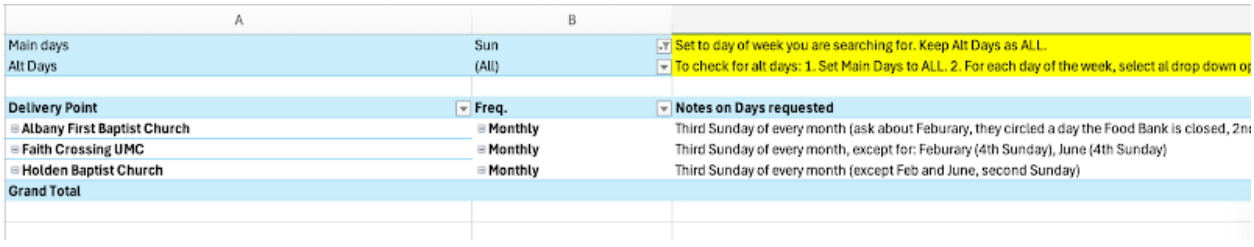
Agency Raw Data Provided by FB | Recorded Agency Requests | Pivot Tool | +

eady Accessibility: Investigate 150%

SOP-001 Figure 1: The Pivot Tool for 2026.

The Pivot Tool can be used to easily sort through the agency's delivery requests. Figure 1 visualizes the Pivot Tool, which is stored in Excel. There are three columns of information, and two filters to organize requests.

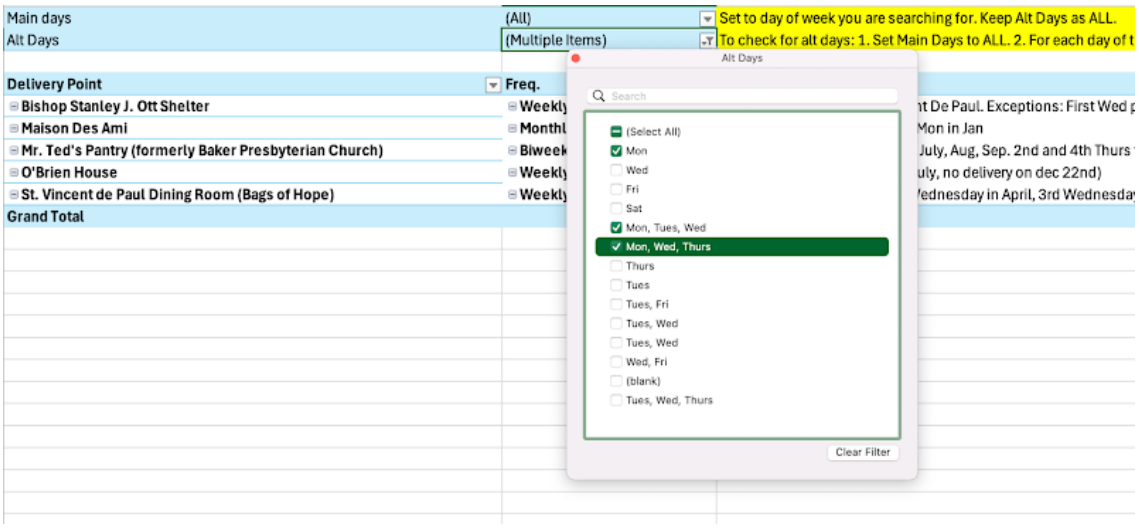
1. Main Days and Alt Days Filters
    a. These filters allow the user to filter requests based on the requested day, for each day of the week. To begin, ensure the Alt Days filter is set to (All). This filter will be used at the end of the process to catch one-off deliveries.
    b. First, open up one blank month of the delivery schedule. Then, select Sunday from the Main Days filter. By entering this selection, all deliveries that have a main delivery day of Sunday each month will appear.



	A	B	
	Main days	Sun	Set to day of week you are searching for. Keep Alt Days as ALL.
	Alt Days	(All)	To check for alt days: 1. Set Main Days to ALL. 2. For each day of the week, select al drop down op
	Delivery Point	Freq.	Notes on Days requested
	Albany First Baptist Church	Monthly	Third Sunday of every month (ask about Feburary, they circled a day the Food Bank is closed, 2n
	Faith Crossing UMC	Monthly	Third Sunday of every month, except for: Feburary (4th Sunday), June (4th Sunday)
	Holden Baptist Church	Monthly	Third Sunday of every month (except Feb and June, second Sunday)
	Grand Total		

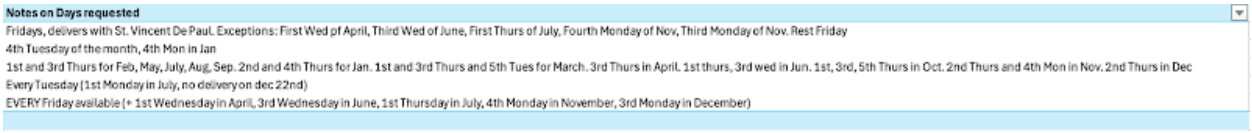
SOP-001 Figure 2: Example of adjusting the Main Days delivery day.

3. By referencing the Notes and Frequency columns, the user can enter the delivery requests within the calendar. Repeat this process for each day of the week, remembering to keep the Alt Day filter set to (All)
4. The use of the Alt Days filter will now be discussed. There are occasions where the agencies make a request that is different from their pattern of deliveries (such as an agency requesting deliveries on Mondays, but then having one delivery on Tuesday due to a holiday.



SOP-001 Figure 3: How to use the Alt Days filter to find one-off deliveries.

5. After recording all main deliveries that occur on Mondays, the user can then check for one-off deliveries that may also occur on Mondays. Set Main Days to (All), and then select any option within the Alt Days drop-down that says "Mon."



SOP-001 Figure 4: The notes column of the Pivot Tool.

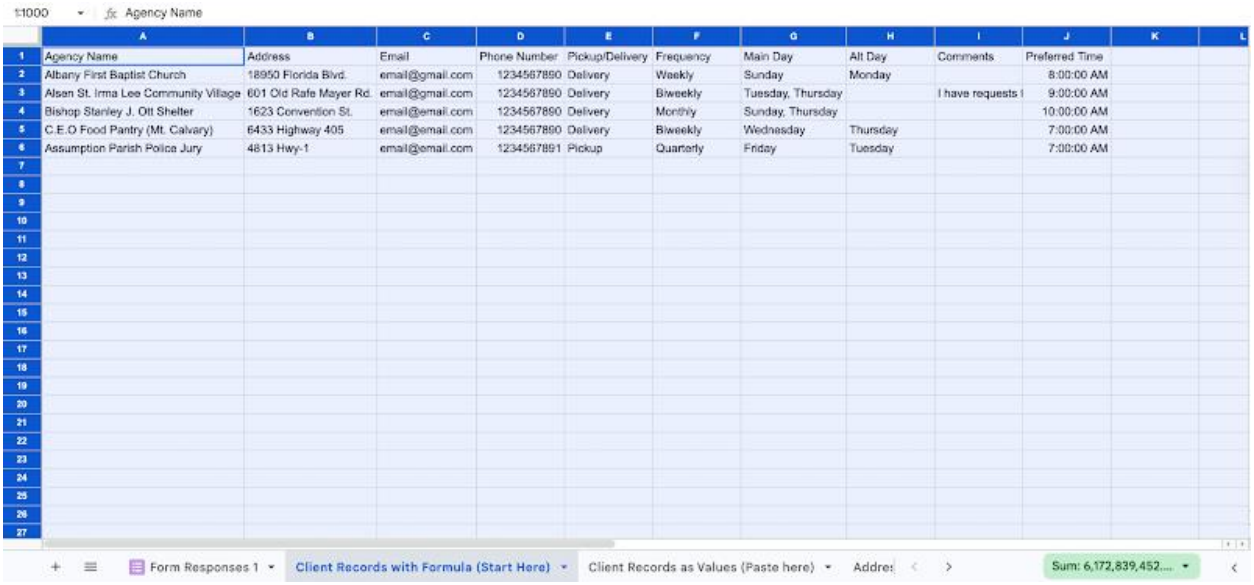
6. Then, it is essential to reference the Notes column in column C. The notes will explain if the shown agencies have a Monday delivery in the specific month you are working on. In this case, the first agency to appear has a Monday delivery in November. If the current month being scheduled were October, this first agency would be ignored.

The Pivot Tool has very detailed notes, and they should be referenced to ensure deliveries are being placed accurately on the calendar.

Part 2: Recreating the Pivot Tool

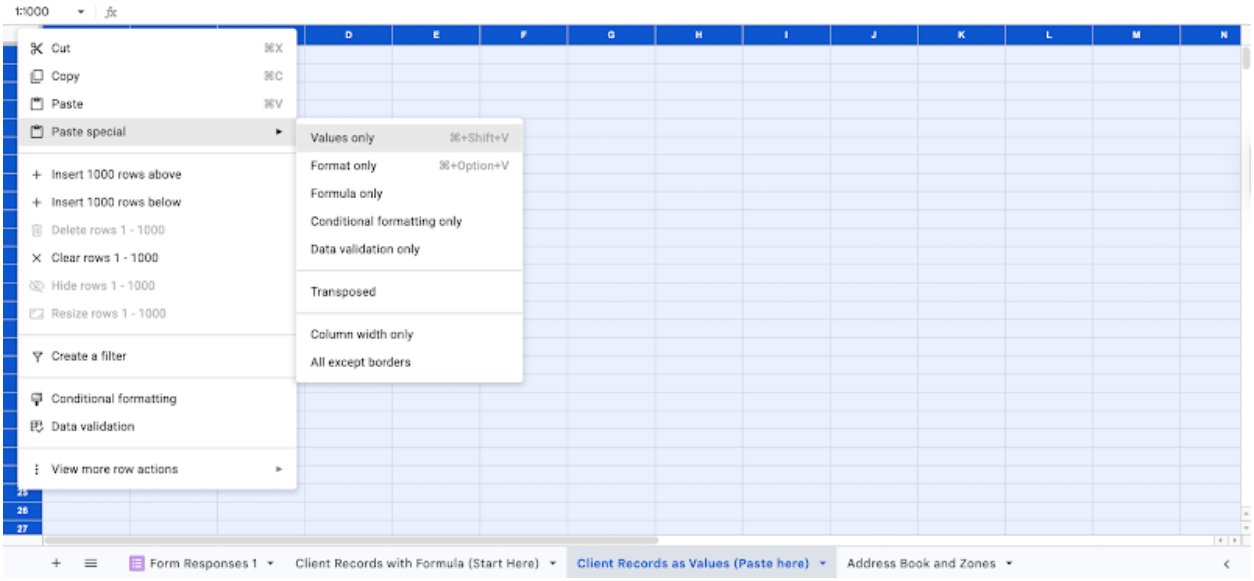
The Google Form for agency data collection, GBRFB Deliveries, gives the user a basis to recreate this Pivot Tool year to year. After having all agencies submit the form (or after internally filling it out for each agency), the entire Client Record Google Sheet will be copied into an Excel Sheet. (The current Google sheet has built-in functions, which will be explained in SOP-002).

This tool has been built on Excel because the pivot tables are easier to use. The Google Sheet simply stores all of the information so that it can be copied en masse to an Excel sheet.



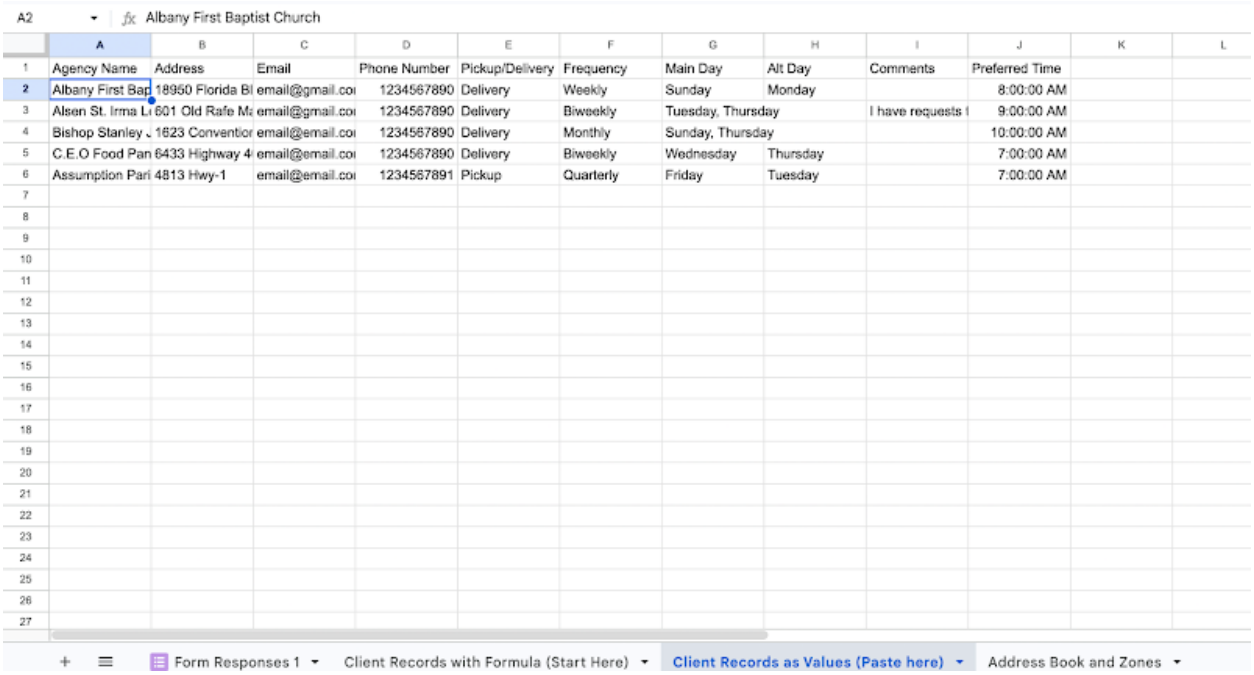
SOP-001 Figure 5: Example of copying the Client Records with Formulas page.

1. Highlight the entire Client Records with Formulas Google Sheet in GBRFB Deliveries (responses). This sheet pulls from the Form Responses sheet, but the values themselves are formulas (IFERROR statements and VLOOKUPS, appendix). To first change these to be actual words (and not formulas) copy the sheet.

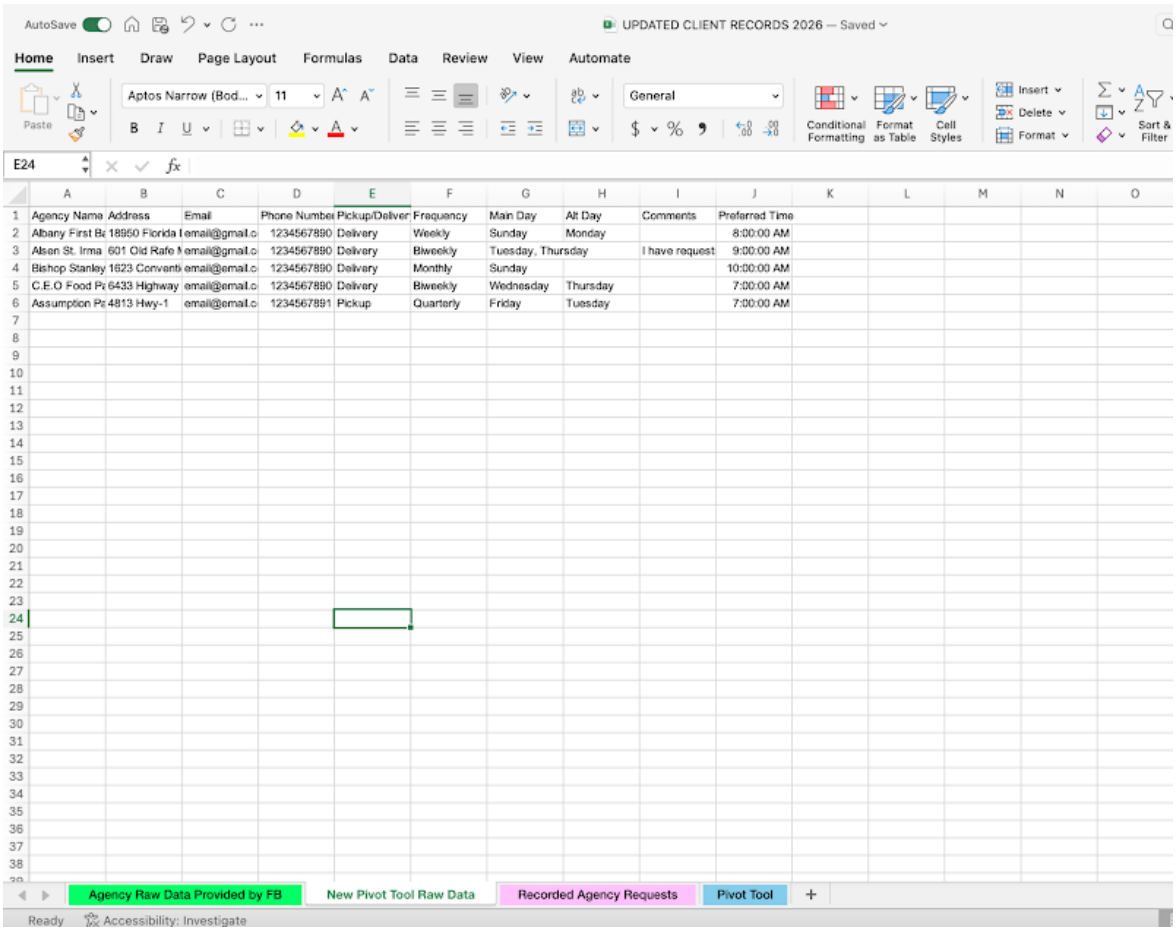


SOP-001 Figure 6: Pasting records over to a new sheet as values, so that formulas are removed.

2. Go to the empty Client Records as Values sheet. Click *Paste Special* and *Values Only*. Now, all of the information will be stored as values, not formulas. At this point, copy this page and paste the information in Excel.



SOP-001 Figure 7: What your new page in Google Sheets will look like. You now copy all information over to Excel.



SOP-001 Figure 8: Excel view of copied information.

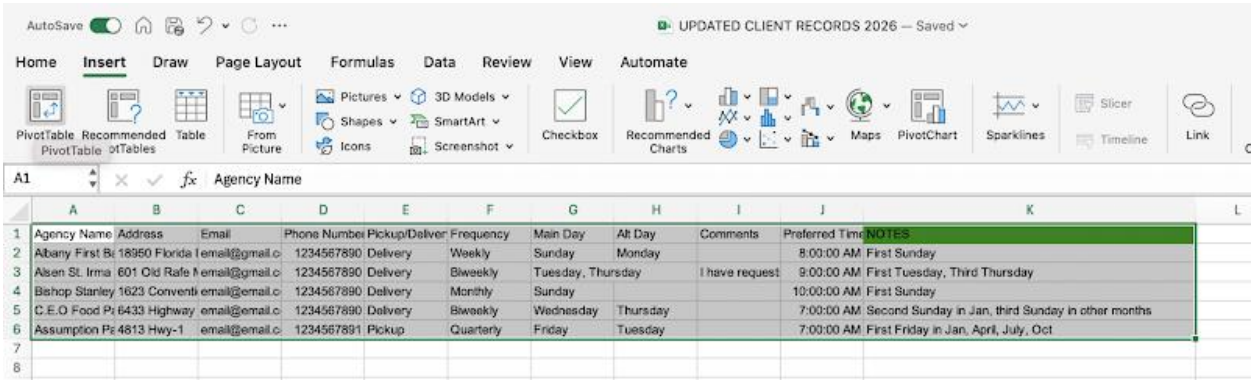
3. Now all information is stored in an Excel sheet. One essential aspect of the pivot table is detailed notes. While this aspect can be submitted on the Google Form through editing the form's questions, it can also be added by hand for every request. At this time, the user can go through the requests and type down notes pertaining to each agency, by referencing their frequency, main and alt days, and the scanned calendar that each agency has submitted (stored in Google Sheet).



	A	B	C	D	E	F	G	H	I	J	K
1	Agency Name	Address	Email	Phone Numbe	Pickup/Deliver	Frequency	Main Day	Alt Day	Comments	Preferred Time	NOTES
2	Albany First Ba	18950 Florida I	email@gmail.c	1234567890	Delivery	Weekly	Sunday	Monday		8:00:00 AM	First Sunday
3	Alsen St. Irma	601 Old Rafe M	email@gmail.c	1234567890	Delivery	Biweekly	Tuesday, Thursday		I have request	9:00:00 AM	First Tuesday, Third Thursday
4	Bishop Stanley	1623 Conventi	email@email.c	1234567890	Delivery	Monthly	Sunday			10:00:00 AM	First Sunday
5	C.E.O Food Pa	6433 Highway	email@email.c	1234567890	Delivery	Biweekly	Wednesday	Thursday		7:00:00 AM	Second Sunday in Jan, third Sunday in other months
6	Assumption Pa	4813 Hwy-1	email@email.c	1234567891	Pickup	Quarterly	Friday	Tuesday		7:00:00 AM	First Friday in Jan, April, July, Oct
7											
8											

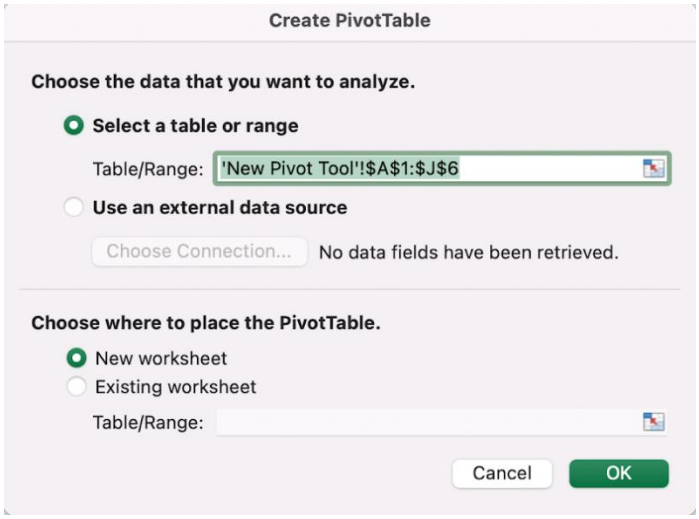
SOP-001 Figure 9: Adding the notes column by hand.

4. Next is creating the Pivot Table. First, highlight all available data. Click *Insert* > *PivotTable.*



SOP-001 Figure 10: Highlighting all information and clicking PivotTable in top left hand corner.

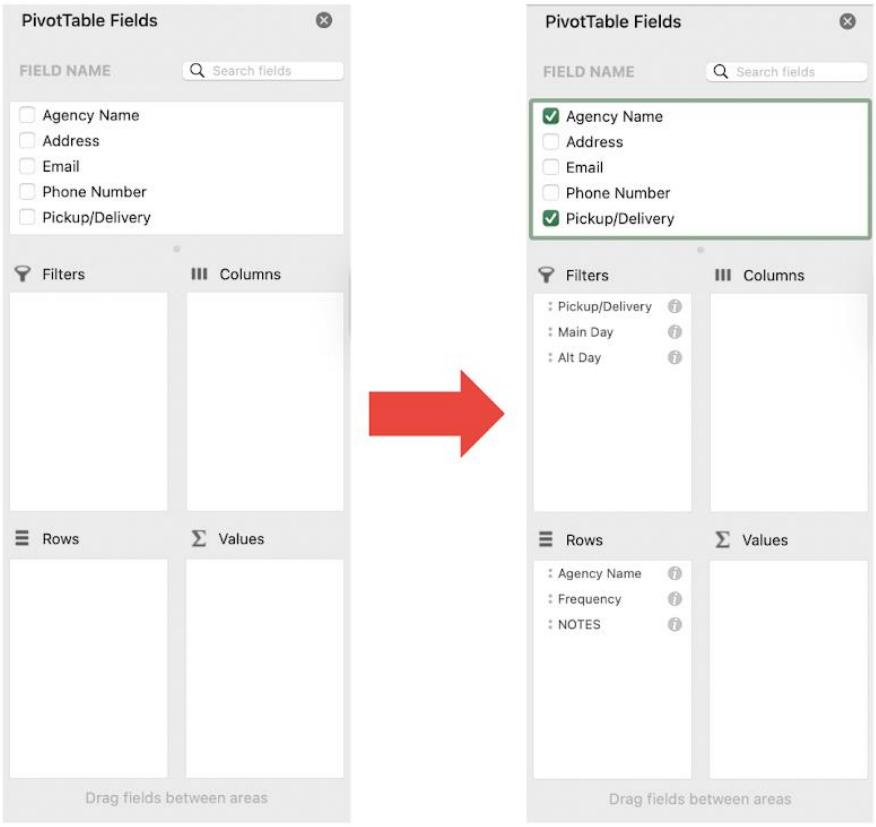
5. The user will be prompted to create the table. Select *OK.*



SOP-001 Figure 11: Prompt for creating the pivot table. This will appear automatically, and the user will select OK.

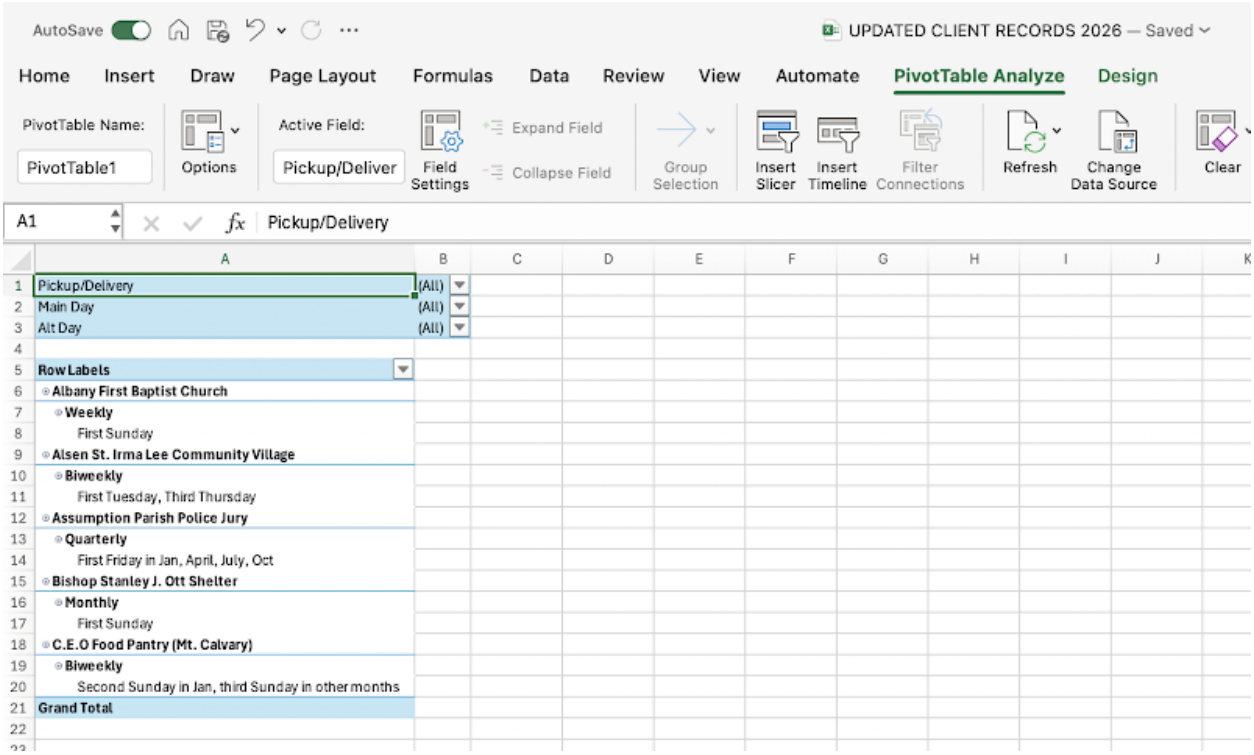
6. The user will now be brought to a new sheet where they can begin dragging fields to create the pivot table. In the side panel that appears, the user will be prompted to drag and

drop fields (which were the columns). Drag and drop the following fields into each respective box.



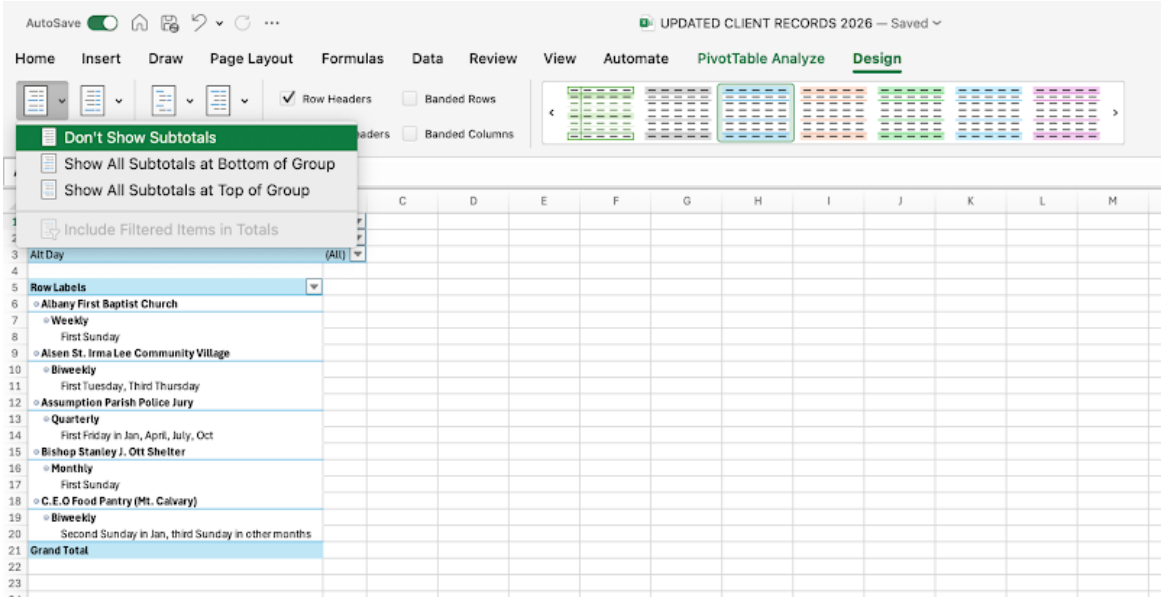
SOP-001 Figure 12: The new panel that appears on the right-hand side of the screen. From here, drag and drop the above fields into each respective space.

1. FILTERS: Fields in this section will appear at the top of the pivot table, and they allow the user to sort by day of the week, alternative days, and pickup or delivery orders.
2. ROWS: Fields in this section will be the data seen within the table.
3. It is possible to add more fields to the rows (such as requested time and agency contact information) if this information is wanted when using the Pivot Tool.

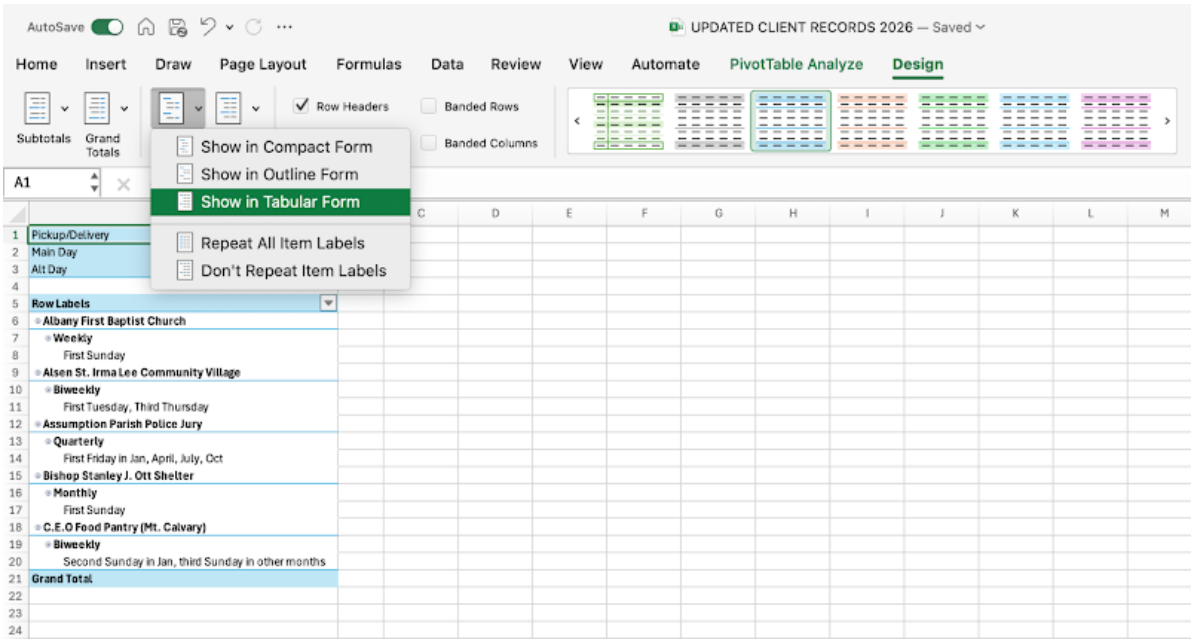


SOP-001 Figure 13: How the pivot table will first look. Two steps must be completed to transform the pivot table into the original tool.

7. At first, the data will appear as it does above. However, the user will click *Design* and follow the prompts below.
1. *Subtotals > Don't Show Subtotals*
2. *Report Layout > Show in Tabular Form*
8. If at any point the screen to edit the table is closed, clicking on the table will bring back up the option to open the *Design* ribbon.

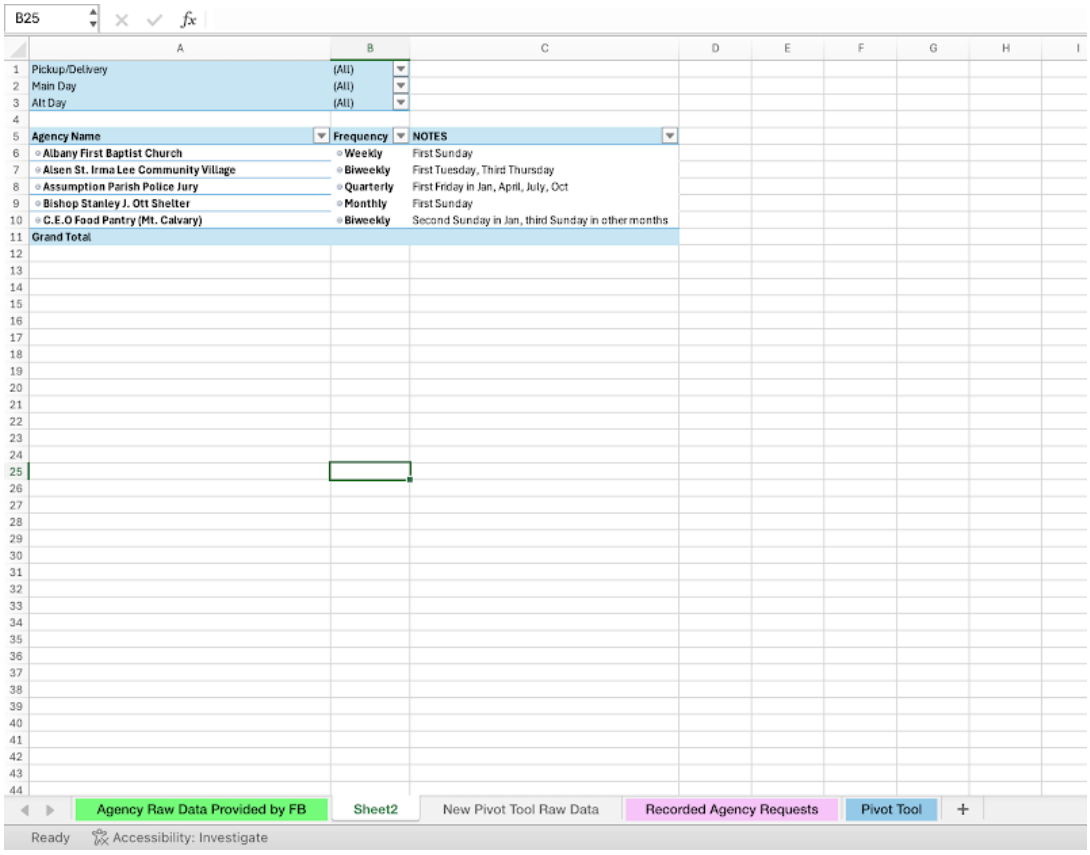


SOP-001 Figure 14: The first step: turning off Subtotals.



SOP-001 Figure 15: Selecting Show in Tabular Form.

9. Now, the pivot table is complete.



SOP-001 Figure 16: Final Pivot Tool.

10. Now, the user can add instructions to the filters, similar to the original tool

Pickup/Delivery	(All)	Set to Delivery
Main Day	(All)	Use this filter to first search for the Main day of deliveries that you are searching for
Alt Day	(All)	Usr this filter last to find any one-off deliveries
Agency Name	**Frequency**	**NOTES**
Albany First Baptist Church	**Weekly**	First Sunday
Alsen St. Irma Lee Community Village	**Biweekly**	First Tuesday, Third Thursday
Assumption Parish Police Jury	**Quarterly**	First Friday in Jan, April, July, Oct
Bishop Stanley J. Ott Shelter	**Monthly**	First Sunday
C.E.O Food Pantry (Mt. Calvary)	**Biweekly**	Second Sunday in Jan, third Sunday in other months
Grand Total		

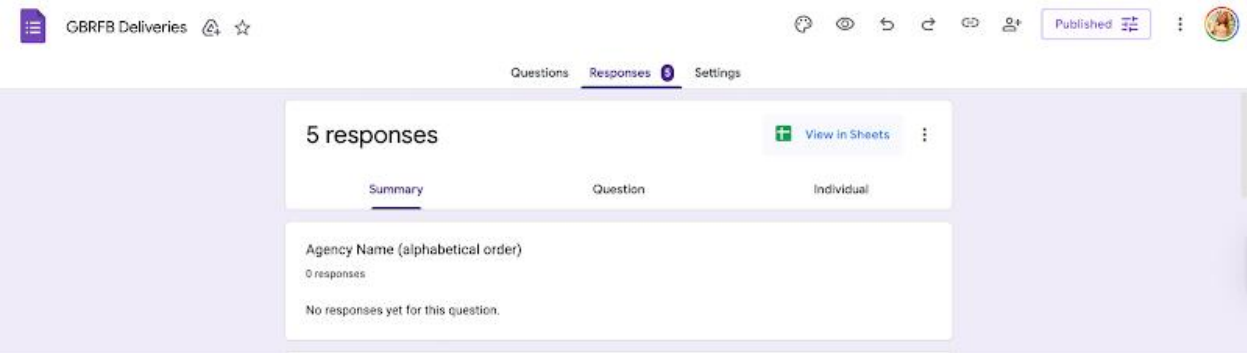
SOP-001 Figure 17: Make adjustments as necessary.

SOP-002: Google Form

[SD]The purpose of this Standard Operating Procedure is to instruct employees on how to edit/adjust the GBRFB Deliveries Google Form. The procedure details how to recreate/edit the delivered *GBRFB Deliveries Form* and *GBRFB Deliveries Responses* Google Sheet. New agencies can be added to the original sheet, and the instructions also explain how to recreate the Google Sheet for later Pivot Table reconstruction (All Pivot Table instructions can be found in SOP-001).

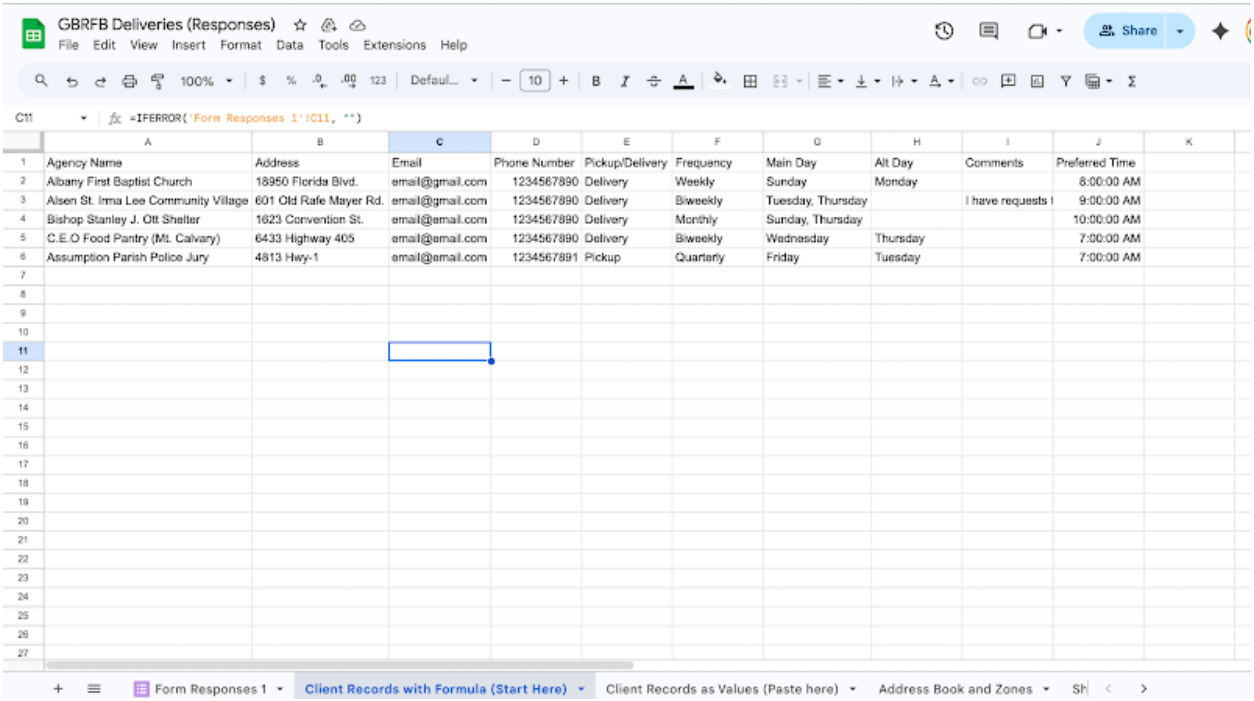
Part 1: Accessing the Response Data

1. The responses to the Google form are first opened in Google Sheets by selecting *View in Sheets*.



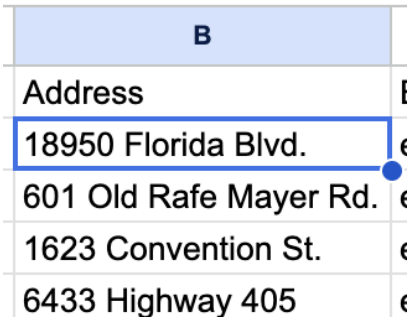
SOP-002 Figure 1: The Google Form response page.

2. After opening Sheets, the responses are found in the *Form Responses 1* Sheet. However, the Sheet you will be referencing is *Client Records with Formula*. This page has specific functions that pull in responses from Form Responses 1 as the form is filled out. The addresses are also pulled from the Address Book and Zones (as agencies are not required to input the address)



SOP-002 Figure 2: Client Records with Formula Sheet.

3. In cell B2, type in =IFERROR(VLOOKUP(A2,'Address Book and Zones'!A:C,3,FALSE), "")
4. This should search for the value in the A column in the Address Book and Zones sheet, and populate the agency's Address Line 1 in the B column.



SOP-002 Figure 3: Cell B2.

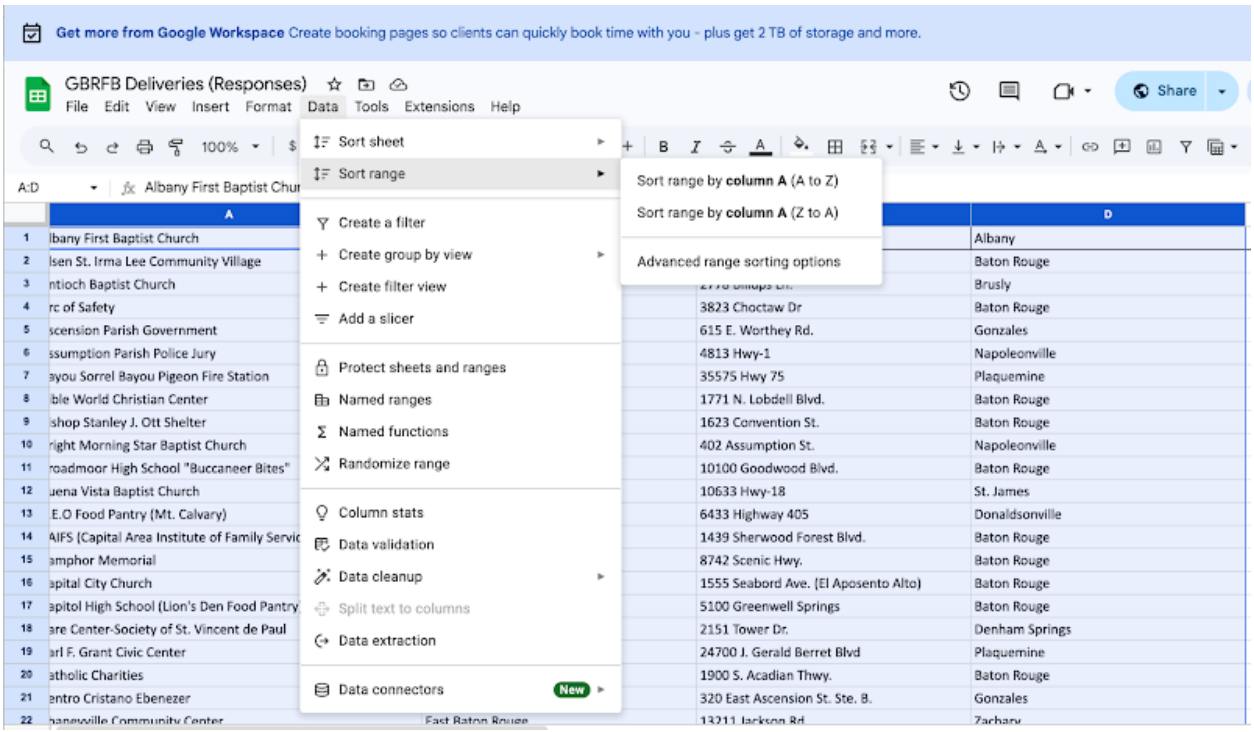
5. Hover over the bottom right corner on cell B2 (where the blue circle is), it should change the cursor into a plus (+). Click this to fill the rest of the column with that formula.

Part 2: Editing the Google Form: Adding Agencies



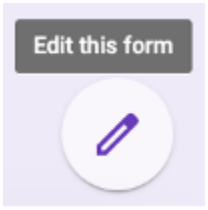
SOP-002 Figure 4: Bottom panel of Workbook.

1. Begin by adding the agency information in the "Address Book and Zones" Tab. This includes the name of the agency, parish, address 1, city, zipcode, latitude/longitude, and zone



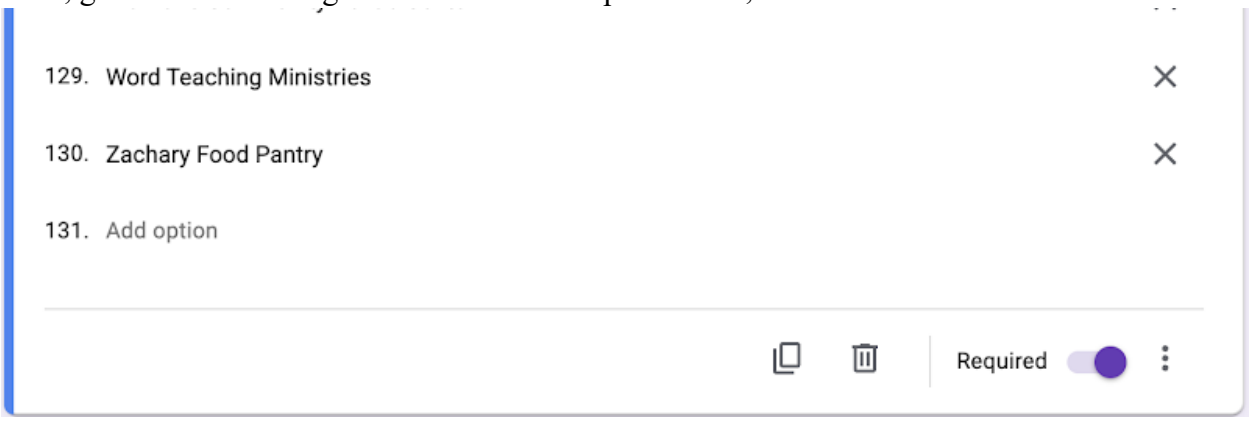
SOP-002 Figure 5: Address Book and Zones Sheet.

2. Highlight the new set of data. Hover over Data, Sort Range, and then click Sort range by Column A (A to Z). This alphabetizes the first column and keeps the row intact.



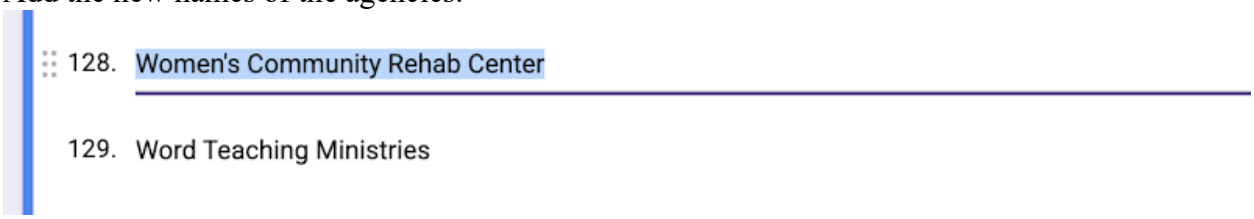
SOP-002 Figure 6: Google Form Editing Icon.

3. Now, go back to the Google Form. Locate this pencil icon, and click it.



SOP-002 Figure 7: Editing Screen for Dropdown.

4. Add the new names of the agencies.



SOP-002 Figure 8: Adjusting Order of Options.

5. To make the options alphabetical, the user can drag the option by clicking and holding down the 6 dots on the left.



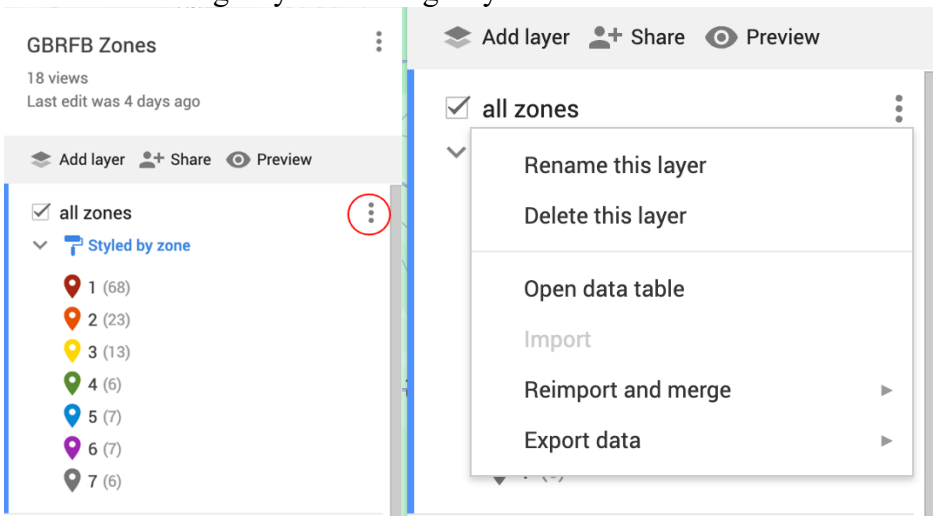
SOP-002 Figure 9: Toolbar.

6. The user can preview the Google Form by clicking on the second icon (the eye) to ensure that it is functional.

SOP-003: MyMaps

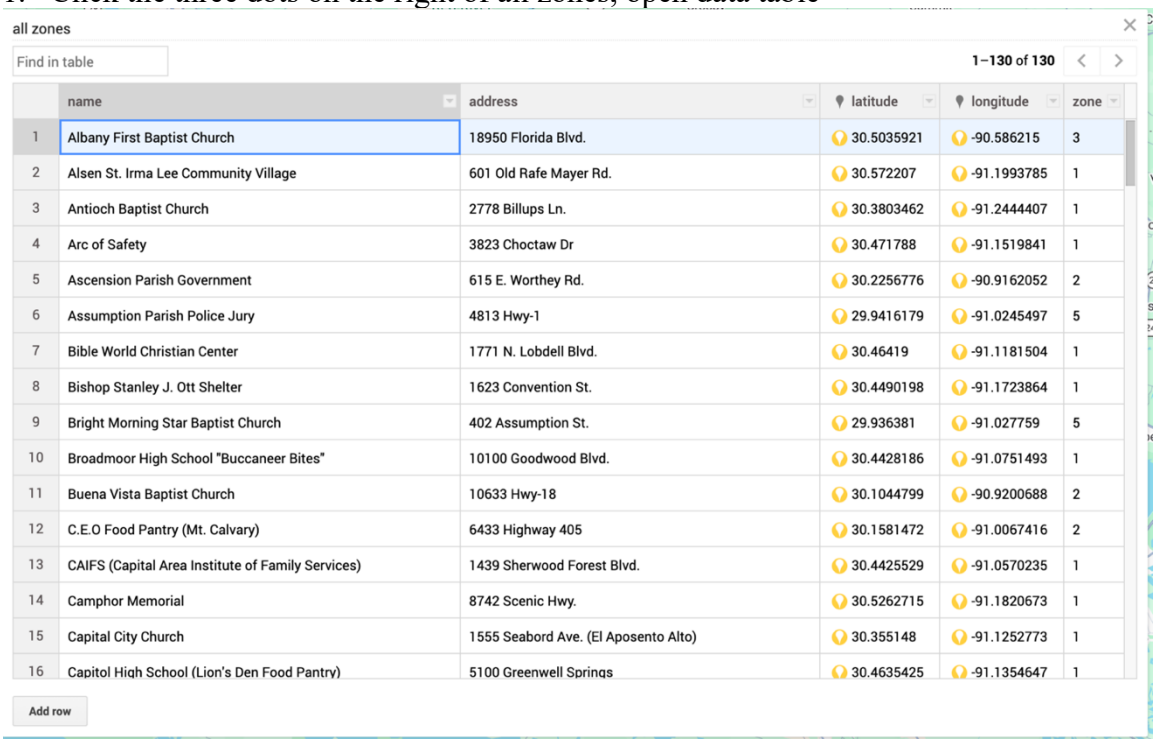
[SD]The purpose of this Standard Operating Procedure is to instruct employees on how to edit the MyMaps tool for accurate agency zoning. Zones are location-based, and by using this tool, the food bank can work towards grouping deliveries by zone to reduce planned mileage.

Background: To add a new agency / edit an agency's zone information:



SOP-003 Figure 1: Sidebar of Main MyMaps.

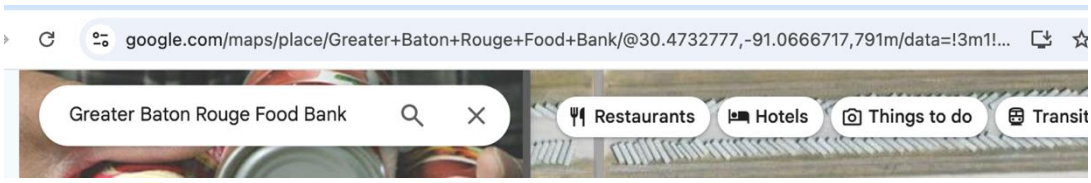
1. Click the three dots on the right of all zones, open data table



	name	address	latitude	longitude	zone
1	Albany First Baptist Church	18950 Florida Blvd.	30.5035921	-90.586215	3
2	Alsen St. Irma Lee Community Village	601 Old Rafe Mayer Rd.	30.572207	-91.1993785	1
3	Antioch Baptist Church	2778 Billups Ln.	30.3803462	-91.2444407	1
4	Arc of Safety	3823 Choctaw Dr	30.471788	-91.1519841	1
5	Ascension Parish Government	615 E. Worthey Rd.	30.2256776	-90.9162052	2
6	Assumption Parish Police Jury	4813 Hwy-1	29.9416179	-91.0245497	5
7	Bible World Christian Center	1771 N. Lobdell Blvd.	30.46419	-91.1181504	1
8	Bishop Stanley J. Ott Shelter	1623 Convention St.	30.4490198	-91.1723864	1
9	Bright Morning Star Baptist Church	402 Assumption St.	29.936381	-91.027759	5
10	Broadmoor High School "Buccaneer Bites"	10100 Goodwood Blvd.	30.4428186	-91.0751493	1
11	Buena Vista Baptist Church	10633 Hwy-18	30.1044799	-90.9200688	2
12	C.E.O Food Pantry (Mt. Calvary)	6433 Highway 405	30.1581472	-91.0067416	2
13	CAIFS (Capital Area Institute of Family Services)	1439 Sherwood Forest Blvd.	30.4425529	-91.0570235	1
14	Camphor Memorial	8742 Scenic Hwy.	30.5262715	-91.1820673	1
15	Capital City Church	1555 Seabord Ave. (El Aposento Alto)	30.355148	-91.1252773	1
16	Capitol High School (Lion's Den Food Pantry)	5100 Greenwell Springs	30.4635425	-91.1354647	1

SOP-003 Figure 2: Data Table Expanded.

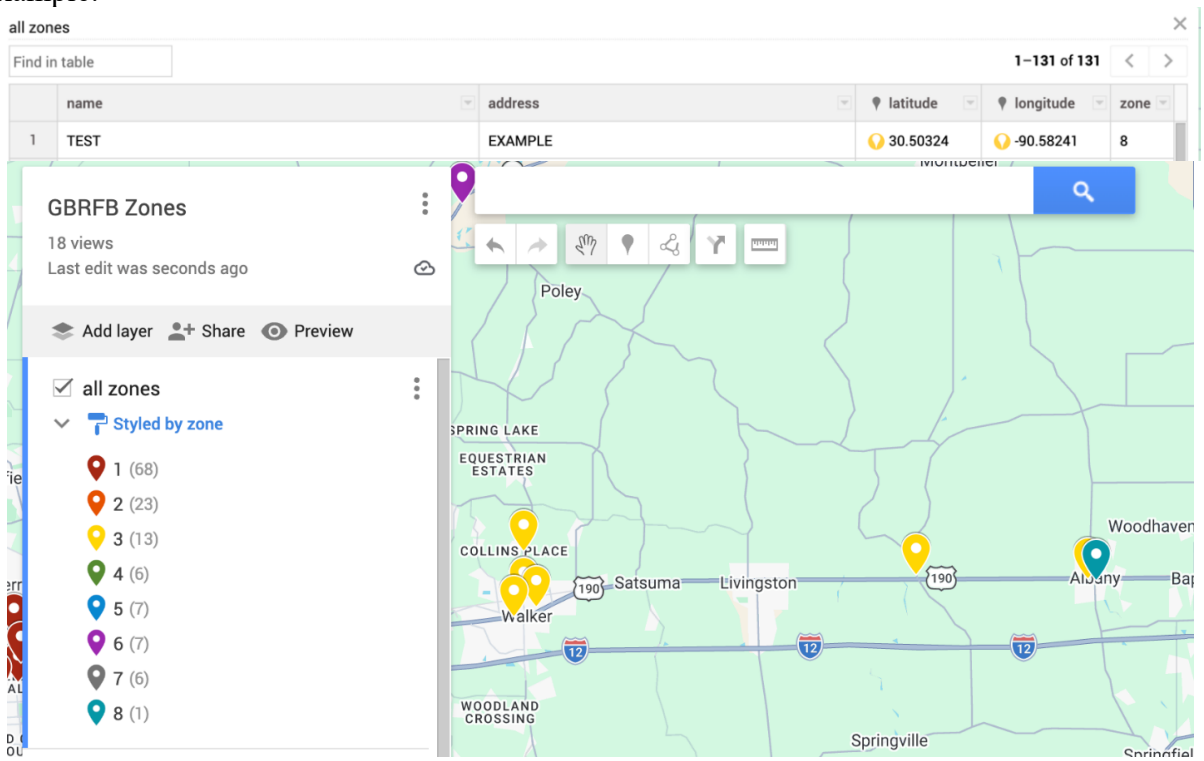
2. Click add row (bottom left), insert the name, address, latitude and longitude (this can be found using google maps, shown below.



SOP-003 Figure 3: Locating Latitude and Longitude of Addresses.

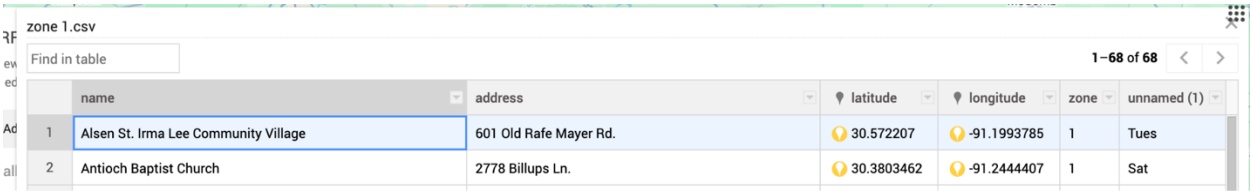
3. For the zone, place a different number to see where this new location would populate, and after seeing which zone it is closest to, change it to the respective number.

Example:



SOP-003 Figure 4: Result after adding new row to data table.

4. The new address is now in zone 3, so reopen the data table, and change the zone value to 3. This will change the marker to yellow.



SOP-003 Figure 5: Data table of individual zone sheet.

5. In the zone 3 layer, repeat the process of opening the data table, adding a row, but instead of changing the zone, the user will add the main requested delivery day (the last column).

SOP-004: Power BI Dashboard

[AS]The purpose of this Standard Operating Procedure is to instruct employees on how to use and recreate the Power BI dashboard.

Part 1: Using Dashboard

1. **Select Truck**
   a. Select the truck of interest in the filter to view data for the specific vehicle.
2. **Click Button**
   b. Click the button to view the operational schedule in Google Sheets or Excel.

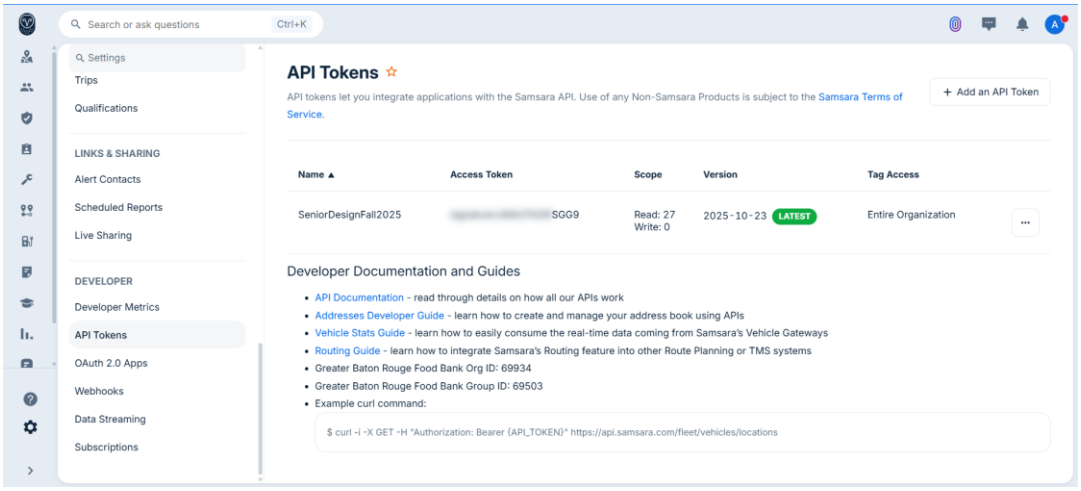
Part 2: Replacing API Token

Updates the API connection so the dashboard continues to pull live data.

To replace the API Token: Follow steps 1-3 → open advanced editor → find the API token section in the advanced editor script and replace it with the new token → click "OK" → Close & Apply.

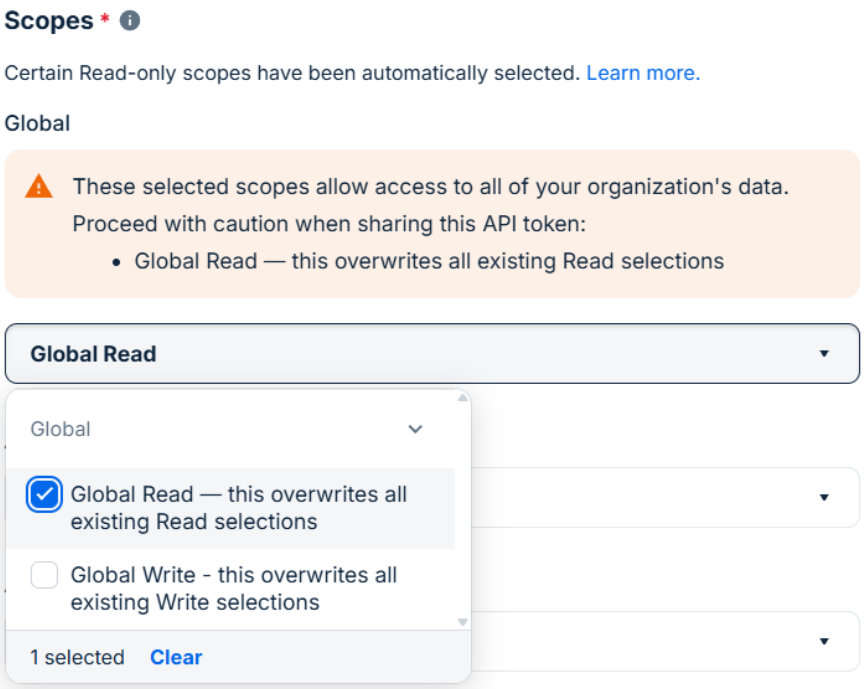
Part 3: Building a new dashboard:

1. In Samsara, go to Settings → API Tokens → Add an API Token.



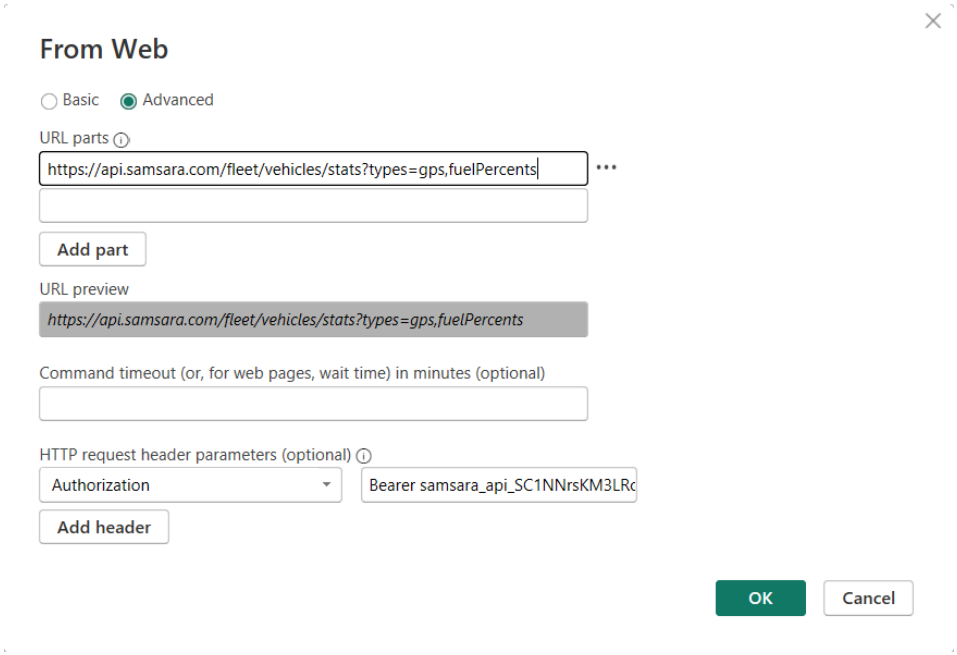
SOP-004 Figure 1: How to generate the API token.

The user should be prompted to allow certain permissions. Enable "Read" for all options. *Grants the necessary read access for retrieving vehicle data.*



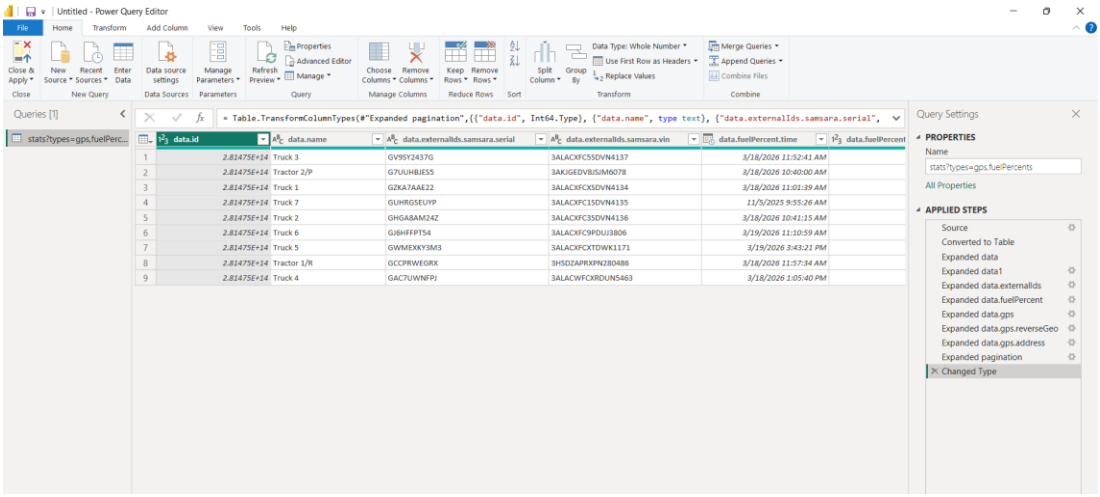
SOP-004 Figure 2: How to enable permissions.

2. Copy the API Token.
3. In Power BI, open Get Data → Web → Advanced. *Starts a new data connection using the Samsara API*
4. Paste "https://api.samsara.com/fleet/vehicles/stats?types=gps,fuelPercents" into the "URL Parts" box. *Specifies the endpoint for retrieving fuel and vehicle data.*
5. In "From Web" window, go to "HTTP request header parameters." Type "Authorization" in first box. In second box, type the word "Bearer", type a space, then paste the API token.



SOP-004 Figure 3: How to paste API token.

6. Click "OK." This should open the Power Query Editor. *Loads the raw API data into Power Query for transformation.*



SOP-004 Figure 4: Power query editor.

7. Name the existing query "Fuel Energy Query."
8. Click New Source → Blank Query → Rename to StartDate
   Enter this script: = #date(Date.Year(DateTime.LocalNow()), 1,1) .
9. Click Home → New Source → Blank Query → Rename to EndDate
   Enter this script: = Date.From(DateTime.LocalNow()) .
10. Click Advanced Editor → insert the following script (replace highlighted section with new token if only interested in replacing API token). *Transforms raw API data into a structured format for analysis. Do not edit script unless experienced in Power BI.*

    let

```
// ----- DATE PARAMETERS (already defined as queries) -----
start  = Date.ToText(StartDate, "yyyy-MM-dd") & "T00:00:00Z",
finish = Date.ToText(EndDate,   "yyyy-MM-dd") & "T23:59:59Z",

// ----- API CALL WITH QUERY PARAMS -----
Source =
   Json.Document(
      Web.Contents(
         "https://api.samsara.com/fleet/reports/vehicles/fuel-energy",
         [
            Query = [
               startDate = start,
               endDate   = finish
            ],
            Headers = [
               Authorization = "Bearer
samsara_api_KP2NUXNzGoxYZwMSZRNWKji84VSGG9",
               Accept       = "application/json"
            ]
         ]
      )
   ),

// ---- EXPAND OUTER RECORD ----
#"Converted to Table1" = Record.ToTable(Source),
Value         = #"Converted to Table1"{0}[Value],
vehicleReports = Value[vehicleReports],

// ---- EXPAND LIST OF VEHICLE REPORTS ----
#"Converted to Table" =
   Table.FromList(
      vehicleReports,
      Splitter.SplitByNothing(),
      null,
      null,
      ExtraValues.Error
   ),

#"Expanded Column1" =
   Table.ExpandRecordColumn(
```

```
        #"Converted to Table",
        "Column1",
        {
            "vehicle", "efficiencyMpgE", "energyUsedKwh", "fuelConsumedMl",
            "distanceTraveledMeters", "estCarbonEmissionsKg", "estFuelEnergyCost",
            "engineRunTimeDurationMs", "engineIdleTimeDurationMs"
        },
        {
            "vehicle", "efficiencyMpgE", "energyUsedKwh", "fuelConsumedMl",
            "distanceTraveledMeters", "estCarbonEmissionsKg", "estFuelEnergyCost",
            "engineRunTimeDurationMs", "engineIdleTimeDurationMs"
        }
    ),

// ---- GET VEHICLE NAME ----
#"Expanded vehicle" =
    Table.ExpandRecordColumn(
        #"Expanded Column1",
        "vehicle",
        {"name"},
        {"VehicleName"}
    ),

// ---- UNIT CONVERSIONS ----

// mL → gallons
#"Added Gallons" =
    Table.AddColumn(
        #"Expanded vehicle",
        "fuelConsumedGallons",
        each [fuelConsumedMl] / 3785.411784,
        type number
    ),

// meters → miles
#"Added Miles" =
    Table.AddColumn(
        #"Added Gallons",
        "distanceTraveledMiles",
        each [distanceTraveledMeters] / 1609.344,
```

```
        type number
    ),

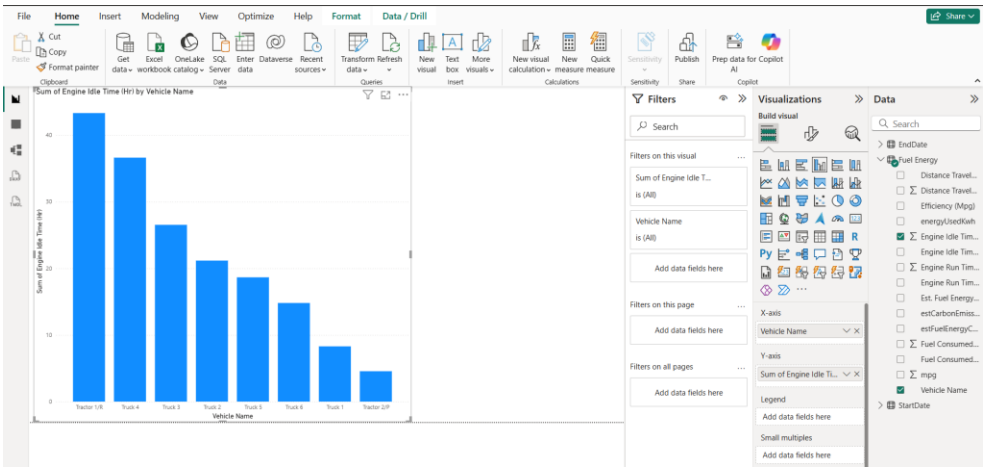
// ms → hours (engine run time)
#"Added Run Hours" =
    Table.AddColumn(
        #"Added Miles",
        "engineRunTimeHours",
        each [engineRunTimeDurationMs] / 1000 / 60 / 60,
        type number
    ),

// ms → hours (idle time)
#"Added Idle Hours" =
    Table.AddColumn(
        #"Added Run Hours",
        "engineIdleTimeHours",
        each [engineIdleTimeDurationMs] / 1000 / 60 / 60,
        type number
    ),

// miles / gallons → MPG
#"Added MPG" =
    Table.AddColumn(
        #"Added Idle Hours",
        "mpg",
        each if [fuelConsumedGallons] <> 0
            then [distanceTraveledMiles] / [fuelConsumedGallons]
            else null,
        type number
    ),
#"Renamed Columns" = Table.RenameColumns(#"Added MPG",{{"efficiencyMpgE",
"Efficiency (Mpg)"}, {"fuelConsumedMl", "Fuel Consumed (Mi)"},
{"distanceTraveledMeters", "Distance Traveled (Meters)"}}),
#"Expanded estFuelEnergyCost" = Table.ExpandRecordColumn(#"Renamed
Columns", "estFuelEnergyCost", {"amount", "currencyCode"},
{"estFuelEnergyCost.amount", "estFuelEnergyCost.currencyCode"}),
#"Renamed Columns1" = Table.RenameColumns(#"Expanded
estFuelEnergyCost",{{"estFuelEnergyCost.amount", "Est. Fuel Energy Cost ($)"},
{"engineRunTimeDurationMs", "Engine Run Time Duration (Ms)"},
```

{"engineIdleTimeDurationMs", "Engine Idle Time Duration (Ms)"},
{"fuelConsumedGallons", "Fuel Consumed (Gal)"}, {"distanceTraveledMiles",
"Distance Traveled (Mi)"}, {"engineRunTimeHours", "Engine Run Time (Hr)"},
{"engineIdleTimeHours", "Engine Idle Time (Hr)"}}),
  #"Filtered Rows" = Table.SelectRows(#"Renamed Columns1", each true),
  #"Renamed Columns2" = Table.RenameColumns(#"Filtered Rows",{{"VehicleName",
"Vehicle Name"}})
in
  #"Renamed Columns2"

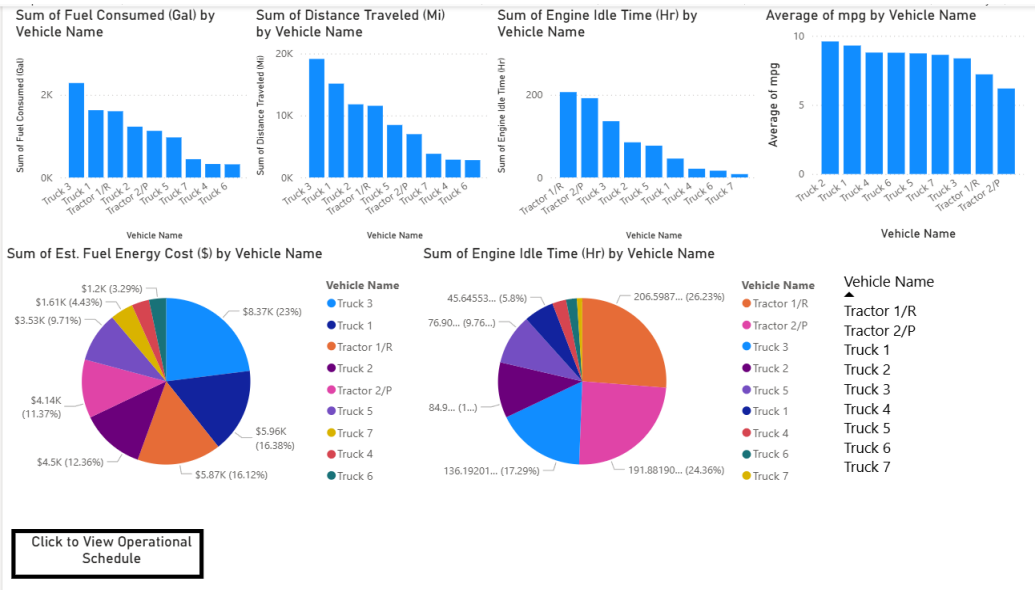
11. Click “Close and Apply.” This will take you to the report view where you can build visuals.
12. Select the visual you want → Drag fields from right side into chart (ex. Vehicle Name to x-axis, mpg to y-axis).



SOP-004 Figure 5: Example of graph that can be created.

If visual looks wrong, go to Data View → EstFuelEnergyCost.amount → Change Data type → Decimal Number → Summarization → Sum. *Ensures values are aggregated correctly in the dashboard.*

13. To add a button to link anything (ex. operational schedule, Samsara), click Insert → Buttons → Blank. In right window, toggle action “on” → For type, select “Web URL”, then paste the web URL of your choice. The button can be customized in the same right window under “Button” and “General.”



SOP-004 Figure 6: Example of final dashboard.

Common Errors:

- API not working → Check token expiration
- Blank data → Check date range
- Authorization error → Confirm “Bearer” format

SOP-005: Driver/ Management Standard Operating Procedure

[MP]The purpose of this SOP is to standardize driver route execution procedures to improve route efficiency, ensure timely food deliveries to partner agencies, and maintain accurate operational data collection through Samsara. This procedure supports route optimization improvements developed through the Senior Design Transportation Improvement Project.

Part 1: Scope

This SOP applies to all drivers operating delivery routes for the Greater Baton Rouge Food Bank.

It covers:

- Pre-trip preparation
- Route execution
- Agency delivery procedures
- GPS/data tracking requirements
- Post-route documentation and reporting

Part 2: Responsibilities and Materials

1. Drivers
    - Follow assigned routes as provided
    - Adhere to scheduled time windows
    - Accurately log delivery completion
    - Report deviations, delays, or issues
2. Transportation Manager
    - Assign optimized routes
    - Monitor GPS tracking and route adherence
    - Review mileage and performance metrics
    - Address recurring operational issues
3. Logistics Team
    - Maintain accurate agency data
    - Ensure routes reflect updated delivery constraints

- Monitor capacity and truck utilization

4. Materials / Tools Required
    - Samsara GPS tracking system
    - Delivery manifest
    - Agency contact list
    - Truck Safety Inspection
    - Mobile communication device
    - Assigned delivery route sheet or digital map

Part 3: 4 Step Procedure on Driver Tasks per Trip

The following list details how drivers should conduct each of their deliveries.

Step 1: Pre-Trip Preparation

- Conduct required vehicle safety inspection
- Review assigned route and agency time windows
- Confirm truck capacity aligns with scheduled deliveries
- Verify GPS tracking system is active

Step 2: Route Execution

- Follow assigned optimized route sequence
- Adhere to scheduled delivery windows
- Contact agency if arrival will exceed scheduled window
- Record any deviations from planned route

Step 3: Agency Delivery

- Verify agency name and delivery type (pickup or drop-off)
- Confirm quantity delivered matches plan
- Obtain required signature or delivery confirmation
- Document any discrepancies

Step 4: Post-Route Completion

- Refuel truck before returning to food bank
- Return vehicle to designated location
- Submit delivery documentation
- Report mileage, delays, or operational issues
- Notify management of recurring scheduling constraints

Part 4: Quality / Compliance Checks

- Mileage comparisons will be conducted between planned vs. actual routes
- Delivery time compliance will be monitored
- Driver feedback surveys will be reviewed post-implementation
- Random ride-alongs may be conducted for process verification
- GPS data (Samsara) will be reviewed for route adherence

Part 5: Records & Documentation

The following records must be maintained:

- Delivery manifests
- Route assignment sheets
- GPS mileage reports
- Driver feedback forms
- Incident reports (if applicable)

Records are stored digitally within the Transportation Department's shared drive and within the GPS tracking system database.

Generative AI Statement

Generative AI was used in this report for brainstorming ideas for project approaches and aided in making paragraphs more concise.